HSPT Exam Simulator

Full-Length Practice Exam 12

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which word is closest in meaning to proximity?

A) height
B) nearness
C) direction
D) distance traveled

2. Which word is closest in meaning to expedite?

A) inspect
B) distribute
C) hasten
D) complicate

3. Which phrase best expresses the meaning of nostalgia?

A) affectionate longing for the past
B) uneasiness about an approaching event
C) satisfaction in another person's success
D) curiosity about an unfamiliar place

4. Which word is most nearly opposite in meaning to skeptical, as in skeptical of an explanation?

A) doubtful
B) convinced
C) questioning
D) uncertain

5. Which word is most nearly opposite in meaning to fluent, as in a fluent delivery of a speech?

A) flowing
B) smooth
C) halting
D) effortless

6. Which word is most nearly opposite in meaning to heterogeneous, as in a heterogeneous mixture?

A) varied
B) diverse
C) mixed
D) uniform

7. Fault is to blame as achievement is to ____.

A) credit
B) delay
C) practice
D) effort

8. Encrypt is to decrypt as compress is to ____.

A) record

B) archive

C) decompress

D) transmit

9. Exile is to country as eviction is to ____.

A) tenant

B) home

C) lease

D) rent

10. Which word belongs with canal, duct, and tunnel as a constructed passage?

A) corridor

B) precipice

C) lagoon

D) meadow

11. Three choices describe being or extending all around something. Which describes crossing something instead?

A) encircle

B) intersect

C) surround

D) envelop

12. Which pair consists of nouns whose standard singular and plural forms are spelled identically?

A) analysis; basis

B) criterion; phenomenon

C) medium; radius

D) series; species

13. A guide gives these complete instructions: "If the flag is not red, use the upper path. If the flag is red, use the lower path unless the bell is sounding; in that case, wait." A traveler hears the bell and correctly waits. What must be true?

A) The flag is not red.

B) The upper path is closed.

C) The lower path is closed.

D) The flag is red.

14. Four flags appear from left to right as J, K, L, M. First, the two middle flags exchange places. Then the entire row is reversed. What is the final left-to-right order?

A) J, L, K, M

B) M, L, K, J

C) M, K, L, J

D) K, J, M, L

15. A viewing booth is acceptable only if it has a seat and at least one of these: a window or a camera feed. Which booth meets these requirements?

A) A booth with a seat and a window but no camera feed

B) A booth with a window and a camera feed but no seat

C) A booth with a seat but neither a window nor a camera feed

D) A booth with a camera feed but neither a seat nor a window

16. Which word is closest in meaning to sentry?

A) passenger

B) customer

C) host

D) guard

17. Which word is closest in meaning to tacit, as in tacit approval?

A) reluctant

B) unspoken

C) public

D) temporary

18. Which phrase best expresses vicarious, as in vicarious excitement?

A) felt only after a long delay

B) deliberately hidden from everyone

C) caused by an unexpected danger

D) experienced through another person

19. Which word is most nearly opposite in meaning to peripheral, as in a peripheral issue?

A) central

B) minor

C) secondary

D) incidental

20. Which word is most nearly opposite in meaning to finite?

A) bounded

B) restricted

C) unlimited

D) measurable

21. Which word is most nearly opposite in meaning to benevolent?

A) charitable

B) kindly

C) malevolent

D) well-meaning

22. Parody is to style as caricature is to ____.

A) appearance

B) ownership

C) distance

D) chronology

23. Bilingual is to languages as ambidextrous is to ____.

A) feet

B) eyes

C) ears

D) hands

24. Aroma is to smell as musical pitch is to ____.

A) sight

B) taste

C) balance

D) hearing

25. Which word belongs with aunt, niece, and cousin as a family relationship?

A) nephew

B) colleague

C) visitor

D) coach

26. Three choices describe ways a liquid can move. Which describes slow burning instead?

A) gush

B) smolder

C) trickle

D) seep

27. Which phrase combines words with directly opposing literal meanings?

A) constant noise

B) distant thunder

C) deafening silence

D) gentle whisper

28. Every visitor to an exhibition chooses at least one postcard. No postcard is chosen by every visitor. Which statement must be true?

A) All visitors choose the same single postcard.

B) Not all visitors choose exactly the same set of postcards.

C) No visitor chooses more than one postcard.

D) Some visitor chooses no postcard at all.

29. Files are sorted first by room, with Elm before Pine, and then by priority within each room, with urgent before ordinary. K is Elm ordinary; L is Pine urgent; M is Elm urgent; N is Pine ordinary. What is the correct order?

A) L, M, K, N

B) M, L, K, N

C) M, K, L, N

D) K, M, N, L

30. Cards R, S, and T are stacked with R at the bottom and T at the top. Remove the top card, add U to the top, remove the top card, and remove the top card again. Which card remains?

A) S

B) R

C) T

D) U

31. Which phrase is closest in meaning to tether in the phrase a dog's tether?

A) restraining line

B) covered shelter

C) trail marker

D) feeding dish

32. Which phrase best expresses relegate, as in relegate someone to a minor role?

A) promote to higher authority

B) ask for an expert opinion

C) assign to a lower position

D) copy a distinctive manner

33. Which phrase is closest in meaning to recant a stated belief?

A) reaffirm it firmly

B) explain it more clearly

C) postpone discussing it

D) withdraw it publicly

34. Which word is most nearly opposite in meaning to convergent, as in convergent paths?

A) divergent

B) meeting

C) approaching

D) joining

35. Which word is most nearly opposite in meaning to legible?

A) bold

B) neat

C) printed

D) unreadable

36. Which word is most nearly opposite in meaning to synchronous, describing events occurring at the same time?

A) simultaneous

B) concurrent

C) staggered

D) coincident

37. Invitation is to guest as invoice is to ____.

A) bystander

B) spectator

C) customer

D) tourist

38. Reference asterisk is to footnote as map symbol is to ____.

A) border

B) legend

C) latitude

D) projection

39. Anonymous is to name as undated is to ____.

A) location

B) signature

C) date

D) price

40. Which word belongs with yesterday, today, and tomorrow as an expression of time?

A) nearby

B) below

C) outside

D) soon

41. What do memorial, tribute, and eulogy have in common?

A) They request forgiveness for an offense.

B) They can honor or remember a person.

C) They formally announce disagreement.

D) They promise a particular future action.

42. Which group forms three standard adverbs by adding only the letters ly, without changing or adding any other letters?

A) calm, brief, stern

B) happy, easy, angry

C) basic, tragic, automatic

D) gentle, simple, humble

43. A copper panel can enter Station C only if it is flat and dry. It can enter Station S only if it has exactly one of a red tag and a green tag. A dry, nonflat panel has both tags. Which preparation satisfies both stations' requirements?

A) Flatten it without changing its tags.

B) Flatten it and remove its green tag.

C) Remove its green tag without flattening it.

D) Flatten it and wet it without changing its tags.

44. Recording R starts before recording S starts and ends after S ends. Recording T begins after S ends but before R ends. What must be true?

A) T begins after R starts and before R ends.

B) T must finish before R ends.

C) R ends before T begins.

D) S and T must overlap in time.

45. A valid label begins and ends with the same letter and contains no pair of identical letters side by side. Which label is valid?

A) KABYA

B) KAAYK

C) KKBYK

D) KABYK

46. Which phrase is closest in meaning to coax?

A) accuse publicly

B) describe precisely

C) depart suddenly

D) persuade gently

47. Which phrase best expresses impetus, as in the impetus for a change?

A) final destination

B) written description

C) driving force

D) unintended obstacle

48. Which phrase best expresses arbitrary, as in an arbitrary decision?

A) required by a clearly stated standard

B) supported by an established calculation

C) reached through a formal agreement

D) made without a consistent reason or rule

49. Which word is most nearly opposite in meaning to callous, describing someone's response to another person's suffering?

A) compassionate

B) indifferent

C) unfeeling

D) insensitive

50. Which word is most nearly opposite in meaning to counterfeit, as in a counterfeit signature?

A) authentic

B) imitated

C) forged

D) fraudulent

51. Which phrase is most nearly opposite in meaning to coercion?

A) compliance obtained through threats

B) obedience secured by force

C) freely given consent

D) agreement extracted under pressure

52. An alibi supports a claim of presence elsewhere, just as a receipt supports a claim of ____.

A) payment merely requested

B) payment already made

C) a price newly increased

D) delivery promised for the future

53. A palindrome retains the same letter sequence after reversal. In a similar relationship, a perfect sphere retains the same appearance after ____ about its center.

A) rotation

B) stretching

C) flattening

D) slicing

54. Intermission is to a play as halftime is to ____.

A) an archery round

B) a marathon run

C) a swimming race

D) a basketball game

55. Which word does NOT name a taste?

A) bitter

B) chilly

C) sour

D) salty

56. Which group consists entirely of words that make a claim less definite?

A) always, certainly, undoubtedly

B) perhaps, possibly, presumably

C) therefore, hence, consequently

D) before, afterward, meanwhile

57. Which pair uses the same verb first for removing a fine material and then for adding one?

A) wash the cloth; wash the plate

B) peel the orange; peel the potato

C) dust the shelf; dust the pastry with sugar

D) polish the spoon; polish the buckle

58. Only three sketches are possible: a crane with curves but no shading, a boat with curves and shading, or a pine with shading but no curves. You learn that the sketch has curves. Which yes-or-no question will now identify it?

A) Does it have curves?

B) Is it either the crane or the boat?

C) Is it the pine?

D) Does it have shading?

59. A grandparent is a parent of someone's parent. Nora and Venn have exactly the same parents. Dara is Venn's parent, and Orr is Dara's parent. What must Orr be to Nora?

A) a parent

B) a child

C) a grandparent

D) a sibling

60. In a contact network, every direct link works both ways. Links through a third person do not establish a direct link. Ena is directly linked to Fenn, and Fenn is directly linked to Goro. Which statement must be true?

A) Ena is directly linked to Goro.

B) Fenn is directly linked to Ena.

C) Ena cannot be directly linked to Goro.

D) Goro has no direct link to Fenn.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. All two-digit positive integers whose tens digit is smaller than their ones digit are listed in increasing order. What immediately follows 29?
..., 26, 27, 28, 29, ____

A) 30

B) 34

C) 32

D) 33

62. For n = 1, 2, 3, and so on, record the least perimeter of a rectangle made from exactly n unit squares arranged in rows, without gaps or overlaps. The first six perimeters are 4, 6, 8, 8, 12, 10. What is the seventh?

A) 16

B) 12

C) 14

D) 28

63. Twelve lamps numbered 1 through 12 stand in a circle. Starting with lamp 1, count only lit lamps, turning off every second one and continuing around. The lamps turned off begin 2, 4, 6, 8, 10, 12, 3, 7. Which is next?

A) 9

B) 1

C) 11

D) 5

64. Begin with the digit string 0. At each stage, copy the entire current string, change every 0 in the copy to 1 and every 1 to 0, and append that copy. The strings begin 0, 01, 0110, 01101001. Which eight digits are appended next?

A) 01101001

B) 10010110

C) 10101010

D) 10011010

65. Given that 37 × 24 = 888, what is 74 × 12?

A) 444

B) 1776

C) 888

D) 3552

66. A positive integer n leaves remainder 4 when divided by 9. What remainder does 2n + 5 leave when divided by 9?

A) 4

B) 0

C) 5

D) 8

67. Two machines start with the same number w, where w > 10. Machine P subtracts 10 and then halves the result. Machine Q halves w and then subtracts 10. How does P's output compare with Q's?

A) P's output is 5 less.

B) The outputs are equal.

C) P's output is 10 greater.

D) P's output is 5 greater.

68. A data set has mean m. Two values, m − 3 and m + 3, are added to the set.
Quantity A: the original mean.
Quantity B: the new mean.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. The numbers a and b satisfy a < b < 0.
Quantity A: 1/b.
Quantity B: 1/a.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. The positive numbers x and y satisfy x + y = 10.
Quantity A: xy.
Quantity B: 24.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. Identical regular tiles meet around one point, with no gaps or overlaps.
Quantity A: the number of equilateral triangular tiles needed at that point.
Quantity B: the number of square tiles needed at that point.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. A solid cube is painted on all six faces and cut into 27 equal small cubes arranged 3 by 3 by 3.
Quantity A: the number of small cubes with no painted face.
Quantity B: the number with exactly three painted faces.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. Four rods of lengths 2, 3, 4, and 5 are joined end to end, in that order, to form a convex quadrilateral. The joints can move.
Quantity A: the interior angle between the rods of lengths 2 and 3.
Quantity B: 90°.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. Each term is formed by subtracting consecutive unit fractions: 1 − 1/2, then 1/2 − 1/3, then 1/3 − 1/4, and so on. What comes next?
1/2, 1/6, 1/12, 1/20, ____

A) 1/24

B) 1/25

C) 1/42

D) 1/30

75. Rows begin and end with 1. Each interior entry of a new row is the sum of the neighboring entries in the previous row. After rows 1; 1, 1; 1, 2, 1; and 1, 3, 3, 1, which row follows?

A) 1, 3, 6, 3, 1

B) 1, 4, 6, 4, 1

C) 1, 4, 4, 4, 1

D) 1, 6, 6, 6, 1

76. Start at 0. At step k, subtract k from the previous term if the result is nonnegative and has not appeared; otherwise add k. The first six steps give 0, 1, 3, 6, 2, 7, 13. What is the next term?

A) 6

B) 7

C) 20

D) 19

77. Positive numbers a and b satisfy a + b = 30 and 1/a + 1/b = 1/6. What is ab?

A) 5

B) 36

C) 90

D) 180

78. For every pair of nonzero real numbers a and b, which expression must equal a/b?

A) (−a)/(−b)

B) (−a)/b

C) b/a

D) −b/a

79. An integer n is greater than 10 and less than 40. It is divisible by 3, leaves remainder 1 when divided by 4, and leaves remainder 3 when divided by 5. What is n?

A) 21

B) 28

C) 33

D) 39

80. If n is any integer, which expression must be odd?

A) $n^2 + n$

B) $n^2 + n + 1$

C) $2n^2$

D) $2n + 2$

81. A positive amount increases by 10% in each of two successive steps.
Quantity A: the overall percent increase.
Quantity B: 20%.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. The positive integers x and y have greatest common factor 1.
Quantity A: the denominator of (x + y)/(xy) when reduced to lowest terms.
Quantity B: xy.

A) Quantity A is greater.

B) Quantity B is greater.

C) The relationship cannot be determined from the information given.

D) The two quantities are equal.

83. Two distinct prime numbers have an even product.
Quantity A: their sum.
Quantity B: 11.

A) Quantity A is greater.

B) Quantity B is greater.

C) The relationship cannot be determined from the information given.

D) The two quantities are equal.

84. An equilateral triangle and a regular hexagon have the same positive perimeter.
Quantity A: one side length of the triangle.
Quantity B: one side length of the hexagon.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. Angle A of triangle ABC is obtuse.
Quantity A: the sum of angles B and C.
Quantity B: 90°.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. A convex heptagon has one exterior turning angle taken at each vertex while traveling once around its boundary in the same direction. Use positive angle measures.
Quantity A: the sum of these turning angles.
Quantity B: 360°.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. The integers 1 through 10 are listed in alphabetical order by their English spellings. What comes next?
8, 5, 4, 9, 1, 7, 6, 10, ____

A) 2

B) 7

C) 9

D) 3

88. Coins have values of 1, 3, and 4 units, with unlimited coins available. The least numbers of coins needed for totals 1 through 7 are 1, 2, 1, 1, 2, 2, 2. Which three terms follow for totals 8, 9, and 10?

A) 2, 3, 3

B) 2, 2, 3

C) 2, 3, 4

D) 3, 3, 3

89. Write 1 through 16 in four rows of four, left to right. Read clockwise around the outside from the top-left corner, then clockwise around the inner four entries from their top-left corner. What comes next?
1, 2, 3, 4, 8, 12, 16, 15, 14, 13, 9, 5, 6, 7, ____

A) 10

B) 11

C) 8

D) 12

90. A sequence consists of runs of 1s and 2s, alternating between the two digits and starting with 1s. The run lengths, read in order, equal the sequence itself. Its beginning is 1, 2, 2, 1, 1, 2, 1, 2, 2, 1, 2, 2. What are the next four digits?

A) 1, 2, 1, 1

B) 1, 2, 2, 1

C) 2, 1, 1, 2

D) 1, 1, 2, 1

91. A two-digit integer has nonzero digits whose product is 48. Its larger digit is in the tens place. What is the integer?

A) 68

B) 64

C) 86

D) 88

92. Three different real numbers have least value 2 and greatest value 14. Add the absolute differences for all three pairs of numbers. What is the sum?

A) 24

B) 12

C) 28

D) 36

93. What is the value of $(1 - 1/2) \times (1 - 1/3) \times (1 - 1/4) \times ... \times (1 - 1/12)$?

A) 1/144

B) 1/12

C) 11/12

D) 1/11

94. The number n is a positive integer written in ordinary decimal notation without leading zeros.
Quantity A: the number of digits in 10n.
Quantity B: one more than the number of digits in n.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. Nonzero real numbers a and b satisfy $a + b < 0$ and $ab > 0$.
Quantity A: a.
Quantity B: 0.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. The real number t satisfies $t^2 < t$.
Quantity A: t.
Quantity B: $1 - t$.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. Triangles ABC and ABD have the same base AB. Points C and D lie on the same line parallel to AB, separate from the line containing AB.
Quantity A: the area of triangle ABC.
Quantity B: the area of triangle ABD.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. Point O is the center of a rectangle with positive side lengths.
Quantity A: the distance from O to a vertex.
Quantity B: the perpendicular distance from O to any one side.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. Two squares each have area 1 square unit. Their corresponding sides are parallel, and their overlap has positive area smaller than 1 square unit.
Quantity A: the overlap area.
Quantity B: 1/2 square unit.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. In a digital font, 0, 6, and 9 each have one enclosed loop, 8 has two, and every other digit has none. Record total loops for consecutive integers starting at 0. The first ten terms are 1, 0, 0, 0, 0, 0, 1, 0, 2, 1. What are the next three terms?

A) 0, 0, 0

B) 1, 1, 1

C) 1, 0, 0

D) 1, 0, 1

101. Replace each pair (a, b) by (b, the remainder when a is divided by b). Which pair fills the gap? (241, 95), (95, 51), (51, 44), ____, (7, 2)

A) (44, 7)

B) (44, 4)

C) (51, 7)

D) (7, 44)

102. A search guesses the lower middle integer of the remaining inclusive range. A low guess removes it and all smaller integers; a high guess removes it and all larger integers. Starting with 1 through 64, the guesses are 32 (low), 48 (high), 40 (high), 36 (low). Which guess comes next?

A) 37

B) 38

C) 39

D) 35

103. In Roman numerals, X = 10, L = 50, I = 1, and V = 5. A smaller symbol immediately before a larger one is subtracted. What does XLIV represent?

A) 46

B) 54

C) 66

D) 44

104. Which expression is equal to $\sqrt{18} + \sqrt{8}$?

A) $5\sqrt{2}$

B) $\sqrt{26}$

C) $\sqrt{10}$

D) $13\sqrt{2}$

105. Two registers read (6, 3). A move subtracts 2 from either register and adds 1 to the other; neither may become negative. After exactly three moves, which pair can occur?

A) (4, 2)

B) (2, 4)

C) (3, 3)

D) (1, 5)

106. How many triples of nonnegative integers (a, b, c) satisfy $a \le b \le c$ and $a + b + c = 5$?

A) 4

B) 5

C) 6

D) 10

107. The number n is real but not an integer.
Quantity A: the smallest integer greater than n.
Quantity B: the greatest integer less than n.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. A list contains five different real numbers. Increase only its greatest number by 5.
Quantity A: the increase in the median.
Quantity B: the increase in the mean.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. Nonzero real numbers a, b, and c satisfy $abc = -1$.
Quantity A: $(ab)(bc)(ca)$.
Quantity B: 1.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. A plane cuts through a sphere of radius 6 units, producing a circular cross section. The plane's distance from the sphere's center is not specified.
Quantity A: the radius of the cross section.
Quantity B: 3 units.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. A triangle and a parallelogram have equal positive areas and equal perpendicular heights.
Quantity A: the triangle's base length.
Quantity B: the parallelogram's base length.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. All four vertices of a rectangle lie on a circle of radius 1 unit.
Quantity A: the rectangle's area.
Quantity B: 1 square unit.

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: After the Shift

I had come to the boat-engine workshop to collect Mom, not to watch her work. I stood beside the doorway with my jacket on and checked the clock above a rack of belts. We were supposed to buy shoes before the shops closed.

"Five minutes," she said without looking up.

She was standing beside an engine suspended over a metal tray. A man in a gray coat pointed at a narrow opening. Mom shook her head, reached for a smaller light, and directed its beam into the gap. Her words were clipped and exact. At home she sometimes forgot what she was saying halfway through a story; here, everyone waited until she finished.

I stopped watching the clock. I watched the others turn when she spoke. One moved a stand before she asked. Another wrote something on a card and brought it over for her to check. I had known that she repaired engines. I had never pictured this small circle of attention around her.

At last she wiped her hands, signed the card, and came toward me. Her jacket was still over one arm when she noticed my collar folded under itself. She straightened it with the hand that had just guided the light.

"Shoes," she said. "Before you grow out of that pair in the doorway."

113. What errand does the narrator expect to do with Mom?

A) collect a repaired engine

B) purchase a workshop clock

C) buy shoes

D) deliver a card to the man in gray

114. Which statement best expresses the passage's central idea?

A) Professional skill requires people to stop caring about their families.

B) Seeing a familiar person in another role can reveal abilities one has never fully noticed.

C) A delayed shopping trip is always more important than completing careful work.

D) Parents can perform a task well only when their children are absent.

115. What does the narrator most likely mean by the "small circle of attention" around Mom?

A) The workers have formed a literal painted circle on the floor.

B) Mom has asked everyone to stop working and admire her clothes.

C) The narrator believes the engine belongs to a theater audience.

D) Her coworkers rely on her judgment and respond closely to her direction.

116. Mom's words at work are described as "clipped." Here, clipped most nearly means ____.

A) brief and sharply delivered

B) physically cut from a sheet

C) too quiet for anyone to hear

D) confused and unfinished

117. Why does the narrator connect Mom's straightening the collar with her guiding the light?

A) The connection suggests that she mistakes the narrator for part of the engine.

B) The connection proves that she can no longer perform ordinary household actions.

C) The linked gestures bring her professional competence and familiar parental care into the same picture.

D) The linked gestures show that the narrator still refuses to notice her work.

Informational Passage: A Wheel That Stores Motion

Some machines receive force in separate pushes rather than in a perfectly steady stream. Without a way to bridge the intervals, their speed would rise during each push and fall between pushes. A flywheel helps reduce this unevenness. It is a wheel designed to store energy while rotating.

When an input speeds the flywheel up, some of the supplied energy becomes stored rotational energy. Between inputs, the turning wheel can give up part of that reserve while helping the connected mechanism continue moving. Its speed decreases as it releases energy, but the variation can be smaller than it would be without the wheel.

How the mass is arranged matters. For the same total mass, placing more of it farther from the axle makes the wheel resist changes in rotational speed more strongly. This resistance does not mean the wheel can never speed up or slow down; it means changing its motion requires more energy transfer.

A flywheel is therefore a temporary store, not a source that produces energy by itself. Friction gradually takes energy from the moving system. If no new input arrives, the wheel eventually slows to a stop. A well-designed flywheel can smooth the delivery of energy, but it cannot remove the need to supply energy in the first place.

118. What is the author's main purpose?

A) To show how a wheel produces unlimited energy without an outside supply

B) To argue that all machines must operate without friction

C) To describe the history of one particular wheel manufacturer

D) To explain how a rotating wheel temporarily stores energy and smooths uneven inputs

119. According to the passage, what happens to the flywheel's speed as it gives up stored energy?

A) It decreases.

B) It increases without limit.

C) It instantly becomes zero.

D) It must reverse direction.

120. For wheels with the same total mass, which arrangement resists changes in rotational speed more strongly?

A) All mass concentrated as close to the axle as possible

B) Mass removed until the wheel has no weight

C) An arrangement that makes the wheel generate energy from nothing

D) More mass placed farther from the axle

121. In the second paragraph, reserve most nearly means ____.

A) an official refusal to participate

B) a seat assigned in advance

C) a supply held for later use

D) an area closed to all machinery

122. Two otherwise identical flywheels begin with the same stored energy. No further input arrives. One loses less energy to friction each second. What is most strongly supported?

A) The wheel with less friction should keep moving longer before stopping.

B) The wheel with less friction must begin creating new energy.

C) Both wheels must stop at the same instant regardless of their losses.

D) The wheel with more friction must retain more of its original energy.

Humanities Passage: A Story in Many Voices

Readers can compare two copies of a printed story word for word. Differences may be treated as printing mistakes. That expectation is less useful for understanding an oral tradition in which each performance is allowed

to vary. Consider the following imagined tradition, whose rules make this distinction clear.
Its central tale concerns a fishing crew that must return before a storm. That return is part of the plot every teller must preserve. Other features are open to invention. The meal on the boat, a passenger's description, or a joke told during the voyage can change for a particular audience. Such embellishment gives the teller room to make the performance immediate.
Listeners who know the tradition judge more than exact wording. They may welcome a new joke while objecting to a change that removes the crew's reason for hurrying. A transcript alone could show which words differ, but understanding those differences requires knowing what the tradition holds fixed and what it permits a teller to reshape.
The common plot is a spine, not a cage. It supports the tale without dictating every movement of its telling. This does not mean that any version is acceptable. It means that continuity can depend on preserving important relationships between events while allowing the language around them to remain alive.

123. In the imagined tradition, which event must every teller preserve?

A) Every passenger eats an identical meal.

B) A particular joke is repeated word for word.

C) The fishing crew returns before a storm.

D) The boat is described with the same adjectives.

124. A researcher finds different jokes in two transcripts of the tale. What should the researcher determine before calling one transcript inaccurate?

A) Whether the differences are permitted variations under the tradition's rules

B) Whether every printed book has exactly the same number of pages

C) Whether the fishing crew used the newest available equipment

D) Whether all the listeners preferred identical meals on boats

125. In the second paragraph, embellishment most nearly means ____.

A) a required correction of the central event

B) an added imaginative detail

C) a word-for-word copy of an earlier performance

D) an explanation of how a printing press works

126. What does "a spine, not a cage" suggest about the common plot?

A) It prevents tellers from making any choices about their words.

B) It provides an organizing structure while leaving room for individual expression.

C) It allows tellers to remove every relationship between events.

D) It is a literal object carried on the fishing boat.

127. Why does the author begin with comparing copies of a printed story?

A) To prove that no printed story can contain a mistake

B) To introduce a familiar standard of sameness before showing why a different standard may fit oral performance

C) To establish that oral tales must be printed before anyone can understand them

D) To begin a chronological account of how fishing boats were invented

128. Which sentence best summarizes the passage?

A) Every difference between two performances proves that one teller has forgotten the story.

B) All listeners judge oral tales only by counting identical adjectives.

C) An oral tradition remains recognizable only when every performer uses completely unrelated events.

D) An oral tradition can preserve a shared plot while allowing variations that follow its own boundaries.

Literary Passage: A Touch to Count

In our fencing drill, the first light tap on a partner's padded vest counted as a touch. We called our own touches when nobody else could see them. My vest had remained untouched all afternoon, and I was beginning to enjoy the clean column beside my name.

Then Jalen stepped in. I felt a small tap below my shoulder just before my own blade reached his vest. My arm had hidden the moment from the coach. Jalen lowered his blade and waited.

"Yours?" the coach asked me.

I could have nodded. It would have been a tiny movement, easier than explaining the sequence. Jalen was already looking toward the next pair, giving me no accusing stare to resist.

The coach's pencil hovered above the page. I looked at the empty column, then at the faint mark on my vest. I could not make what I had felt disappear simply by choosing a different answer.

"His first," I said.

The pencil moved. Jalen raised his blade again and gave me the same quick nod he had given before every other exchange. Nothing in his manner announced that I had done something remarkable.

I pulled my mask back into place. It had not lost any metal, but it sat more lightly on my head.

129. Why does the coach ask the narrator which touch came first?

A) The coach has left the practice room.

B) The narrator's arm hid the moment from the coach.

C) Jalen has refused to wear a vest.

D) The narrator has asked to stop the entire drill.

130. Which statement best expresses the passage's central thought?

A) An unseen touch can never count in a practice drill.

B) Keeping a perfect record is more important than reporting accurately.

C) Honest actions are worthwhile only when they receive public praise.

D) An honest account can matter more than preserving an impressive record.

131. What does Jalen's ordinary nod after the answer most strongly suggest?

A) He believes the narrator has described the sequence incorrectly.

B) He treats truthful reporting as a normal part of the drill.

C) He intends to demand a reward for accepting the answer.

D) He has decided that no further exchanges should take place.

132. What is the effect of the pencil that "hovered above the page"?

A) It shows that the coach has forgotten how to write.

B) It proves that the score sheet is physically out of reach.

C) It holds attention on the brief pause in which the narrator must choose an answer.

D) It shifts the scene into a long account of how pencils are manufactured.

133. What does the final description of the mask sitting "more lightly" convey?

A) The narrator feels relief from the burden of possibly concealing the truth.

B) The mask has actually become smaller during the exchange.

C) The narrator has secretly removed a required piece of equipment.

D) The coach has changed the rules so that touches no longer count.

Informational Passage: A Brown Surface

A freshly cut apple may develop a brown surface even though it was pale a short time earlier. This change depends on a chemical reaction, not simply on the cut surface becoming dry. The apple contains enzymes, proteins that help particular reactions occur, as well as compounds those enzymes can act on.

Within intact tissue, cell structures keep some of these materials separated. Cutting damages the structures and exposes the tissue to oxygen from the air. Materials that were separated can now meet, and oxygen can participate in the reactions that lead to brown pigments. The visible color therefore reflects several requirements coming together.

Different conditions can affect different requirements. A coating that limits oxygen reaching the surface acts as a barrier. It need not remove the enzymes to reduce the reaction. Cooler conditions can slow reaction rates, while a more acidic environment can reduce the activity of the enzymes involved. These approaches act in different ways rather than making every part of the process disappear.

Their effects also have limits. If a barrier is broken and fresh air reaches tissue that still contains the necessary materials, browning can resume. Explaining the change means following the interaction among tissue damage, available oxygen, and enzyme activity, rather than treating brown color as the result of one isolated cause.

134. What does the passage say cutting does to the apple's cell structures?

A) It creates all the enzymes from materials outside the apple.

B) It removes every compound that could form a pigment.

C) It damages structures that kept some materials separated.

D) It prevents the cut tissue from meeting air.

135. Which substance from the air participates in the browning reactions described?

A) nitrogen

B) oxygen

C) argon

D) water vapor

136. What does the author primarily seek to explain about the apple's changing color?

A) To argue that drying is the only process that changes a cut apple's color

B) To describe the history of commercial apple shipping

C) To explain how several interacting conditions produce browning and how changing those conditions can affect it

D) To show that every method of reducing browning permanently removes all enzymes

137. In the third paragraph, barrier most nearly means ____.

A) a substance that creates unlimited oxygen

B) a signal announcing the exact reaction time

C) a measurement of the apple's total weight

D) something that blocks or limits passage

138. A newly coated cut surface remains pale. Which conclusion would go beyond the evidence described in the passage?

A) The pale result proves that the apple's enzymes have been destroyed.

B) The coating may be limiting oxygen reaching the surface.

C) The enzymes may still be present in the tissue.

D) A change in one requirement may be affecting the visible reaction.

Informational Passage: The Cost of an Empty Seat

Consider a simplified ferry service with one sailing already scheduled. For that sailing, the crew receives the same pay and the dock charges the same fee whether the boat carries many passengers or few. These are fixed costs for this decision. Printing an additional ticket, by contrast, adds a small cost when another passenger boards; that is a variable cost.

An empty seat does not make the crew's pay disappear. When fewer passengers buy tickets, the fixed costs are

spread among fewer people. The average cost per passenger can therefore rise even if the total cost of the sailing changes very little.

Average cost is not the same as the extra cost of serving one more passenger. Suppose the scheduled sailing is currently losing money. An additional fare could exceed the extra ticket-printing cost without covering a full share of the crew's pay. The difference would contribute toward fixed costs, though the sailing might remain unprofitable.

The decision's boundary matters. Operating a second boat requires another crew and another dock fee in this model. A cost fixed while filling an existing boat can increase when another sailing is added. Calling a cost fixed is therefore useful only after the decision has been defined; it does not mean that the cost can never change.

139. What is the author's main purpose?

A) To recommend that every passenger always travel without paying

B) To show that an empty seat removes the need to pay the crew

C) To prove that average cost and extra cost are always identical

D) To distinguish costs of an existing sailing from costs that change when the service expands

140. In the model, which cost stays the same when one more passenger boards the already scheduled boat?

A) the cost of printing the additional ticket

B) the number of tickets that must be printed

C) the fare paid by the additional passenger

D) the dock fee for that sailing

141. Why would income sufficient to make one extra passenger worthwhile on an existing sailing not necessarily make a second boat worthwhile?

A) The second boat introduces additional crew and dock costs that the existing-seat decision does not add.

B) A second boat eliminates every variable cost from the service.

C) A fare can never contribute toward any fixed cost.

D) The second boat must have exactly the same number of passengers as the first.

142. When fixed costs are "spread among fewer people," spread most nearly means ____.

A) physically stretched around

B) hidden from

C) allocated across

D) eliminated for

143. How does the final paragraph develop the discussion?

A) It changes the decision from filling an existing boat to adding a sailing, showing why the cost boundary matters.

B) It abandons the ferry example to tell an unrelated personal story.

C) It repeats the claim that all costs stay fixed under every possible decision.

D) It reverses the earlier statement that printing an additional ticket has a cost.

144. Which description best fits the passage's tone?

A) Nostalgic, because it longs for an earlier age of ferry travel

B) Analytical, because it distinguishes costs under explicitly stated assumptions

C) Alarmist, because it predicts an unavoidable disaster on every sailing

D) Triumphant, because it celebrates unlimited income without any expenses

Literary Passage: The Rented Voice

At Pell's stall, Sada rented a golden badge that made its wearer's voice carry across the entire market. She had spent the morning repeating herself to distracted sellers. Now, she thought, nobody would overlook her.
She clipped the badge to her coat and asked the price of a pear. Her words rolled over the stalls. Heads turned, just as Pell had promised. The fruit seller answered quickly, and the people nearest Sada receded a few steps.
Near the fountain, Orin caught her sleeve. He led her behind a stall where hanging rugs screened them from the crowd.
"I need to tell you something," he said. "But not in front of everyone."
"Of course," Sada replied. The badge sent those two words over the rugs and out across the square. Several faces appeared around the stall's corner. Orin released her sleeve and pressed his lips together.
Sada reached for the badge. It had done exactly what the seller said it would do. She had wanted the entire market to hear her, and now the entire market was waiting for another sentence.
She unclipped the gold circle and slipped it into her pocket.
"I'm listening," she said in her ordinary voice.
For the first time that morning, she did not look to see how many heads had turned.

145. What must Sada do to make the rented badge amplify her voice?

A) Place it behind the hanging rugs.

B) Hand it to the fruit seller.

C) Hold it beneath the fountain water.

D) Clip it to her clothing.

146. Which statement best expresses the central thought of Sada's experience?

A) A powerful voice always makes a private conversation easier.

B) Being heard by everyone can interfere with communicating appropriately with one person.

C) A promise becomes false whenever its results disappoint the buyer.

D) Every conversation should attract as many listeners as possible.

147. Why does Orin most likely release Sada's sleeve and press his lips together?

A) Her amplified reply has drawn attention to a conversation he wanted to keep private.

B) He has discovered that the golden badge cannot amplify voices.

C) He wants the fruit seller to announce his secret for him.

D) He has already told the crowd everything he came to say.

148. When the people near Sada receded, what did they do?

A) leaned closer

B) moved back

C) laughed aloud

D) began bargaining

149. What is the effect of saying that the badge did "exactly what the seller said it would do" just before Sada removes it?

A) It reveals that Pell secretly replaced the badge with a powerless copy.

B) It establishes that the badge has stopped working because the rugs block it.

C) It interrupts the story with an unrelated explanation of how gold is mined.

D) It emphasizes the irony that fulfilling the promise has created the opposite of the connection she now needs.

Literary Passage: The Knocking in the Tent

The tapping began after my cousin Aven fell asleep. Three quick knocks, a pause, then two more. Beyond the tent wall, the darkness seemed to be waiting for me to answer.
I had insisted on taking the side nearest the entrance. Earlier, that had meant an easy way out to watch the

sunset. Now it meant being closest to whatever had found us.
The knocks stopped. I held my breath. When they began again, they sounded lower down, near the opening where my boots stood. I lifted the lantern but kept its light pointed at my blanket, as though the brightness might announce where I was.
Aven stirred. "Wind," she murmured, and turned over.
That seemed too small a word for what I had been hearing. I brought the lantern closer to the entrance. In its light, the loose metal zipper pull swung against the water flask I had left beside my boot. A gust pushed the tent fabric inward. Tap, tap, tap.
I tucked the pull into the boot and moved the flask. The next gust made only a soft flutter.
I lowered the lantern. Outside, the same dark branches moved against the same dark sky. They had not become smaller or farther away, but I no longer needed to imagine something hiding between them.

150. What actually makes the tapping sound outside the narrator's blanket?

A) Aven knocks on the tent floor with her hand.

B) A branch repeatedly hits the lantern.

C) The narrator's boot falls against the tent pole.

D) The loose zipper pull strikes the water flask.

151. Which statement best describes the movement of the narrator's experience?

A) Fear gives an uncertain sound a threatening meaning until observation supplies a simpler explanation.

B) A familiar sound becomes more dangerous each time its source is examined.

C) The narrator discovers that Aven has deliberately created an imaginary intruder.

D) The nighttime landscape changes physically once the lantern is extinguished.

152. What does the narrator's decision to point the lantern at the blanket most strongly suggest?

A) The narrator has already recognized the zipper pull as the source.

B) The narrator wants to prevent Aven from seeing the sunset.

C) The narrator fears that showing a light could reveal a position to a supposed intruder.

D) The narrator believes the flask needs light in order to make a sound.

153. In "I tucked the pull into the boot," tucked most nearly means ____.

A) pulled apart with force

B) measured against an edge

C) put into a sheltered place

D) painted a different color

154. Why is Aven's one-word explanation placed before the narrator sees the source of the noise?

A) It proves that Aven has seen every object outside the tent while asleep.

B) It gives a simple possibility that the frightened narrator initially dismisses, delaying the reader's confirmation.

C) It changes the narration from a personal account into a weather report.

D) It shows that the narrator has forgotten there is any wind outside.

Informational Passage: A Pole on Each Piece

A bar magnet has a north pole at one end and a south pole at the other. The names describe magnetic behavior: like poles repel, while unlike poles attract. Cutting the bar between its ends might seem to offer a way to obtain a north-only piece and a south-only piece. That is not what happens. Each piece has both a north pole and a south pole.
A useful model looks inside the material. In an iron magnet, tiny regions called domains behave like small magnets. Within a domain, magnetic effects point in a common direction. When many domains are aligned, their effects reinforce one another and produce the magnet's larger overall effect.

Cutting the bar divides the material, but it does not sort north poles into one pile and south poles into another. In the simplified model, the aligned regions remaining in each piece still produce a magnet with two poles. If the pieces stay in their original orientation, the newly exposed faces have opposite poles and attract one another. The model also distinguishes a region's behavior from the whole object's behavior. In an unmagnetized piece of iron, domains can point in different directions, largely canceling their effects. A weak overall effect therefore need not mean that the small regions themselves have stopped behaving magnetically.

155. Which outcome does the passage explicitly describe after the bar magnet is cut?

A) Each piece has only the kind of pole nearest its original end.

B) All the domains disappear, leaving two pieces with no magnetic behavior.

C) Each piece retains both kinds of pole, and the new faces attract when kept in their original orientation.

D) Both newly exposed faces become north poles and therefore attract.

156. What is the author's main purpose in introducing the domain model?

A) To show that a magnet's north pole is a geographical location inside the material

B) To argue that the smaller a magnet becomes, the fewer kinds of pole it has

C) To give instructions for sorting domains into north-only and south-only collections

D) To explain why dividing a magnet does not separate its two kinds of pole into different pieces

157. Two newly cut faces attract while the pieces remain oriented as before. According to the stated rules, what would happen if one piece were turned end for end so its original outer end faced the other piece's cut face?

A) The facing ends would repel because they would now be like poles.

B) The facing ends would attract more strongly because both would be north poles.

C) The turned piece would lose one of its two poles.

D) Neither piece could exert any magnetic effect after being turned.

158. How do the opening and middle paragraphs work together?

A) They present a historical timeline of increasingly expensive cutting tools.

B) They raise an apparently plausible prediction, reject it, and use a model of smaller regions to explain the actual outcome.

C) They compare two travelers whose routes happen to be called north and south.

D) They state a cutting result and later withdraw it without offering an explanation.

159. Which summary best preserves the final paragraph's distinction between domains and an entire object?

A) Magnetic regions can remain active while their differing directions reduce the object's overall effect.

B) A weak overall effect proves that every domain has lost all magnetic behavior.

C) Every object with domains must have the same overall strength regardless of their directions.

D) Domains exist only at the two outer ends and vanish everywhere else.

Literary Passage: The Button Above the Curtain

For three winters I had kept a coat closed. Wind pressed against my brass face, and the buttonhole held me firmly by the waist. The other buttons and I stood in a straight line. We took our work seriously.

Then the coat was remade. "Too heavy for the new cloth," someone said. Scissors slipped beneath my thread, and I was laid in a dish beside a broken bead. Without a buttonhole to hold, I could think of nothing useful to do.

Rin found me there. She turned me over, tested the holes with a thread, and carried me to a cardboard theater on the table. Two small curtains hung across its opening.

"The moon," she announced.

I would have explained that I was a coat button, if brass had been able to answer. Instead, she passed a string through my holes and held me aloft against a blue paper sky. She hid the string behind a painted cloud, leaving my round face visible. A lamp beside the stage made it shine.
Below me, a paper ship waited on a paper sea. Rin moved it gently from side to side. I had faced real rain without much interest, but I watched that tiny ship as though its crossing mattered.
The curtains parted. After years of keeping something closed, I was waiting for the story to open.

160. Why is the narrator removed from the coat?

A) The button has lost all its holes.

B) Rin has broken the button into pieces.

C) The button is too heavy for the new cloth.

D) The coat needs a larger row of brass buttons.

161. When Rin holds the button aloft, she holds it ____.

A) flat inside the dish

B) tightly beneath the ship

C) high in the air

D) loosely against the floor

162. Why does Rin most likely hide the string behind a painted cloud while leaving the button visible?

A) She wants viewers to see the moon without noticing how it is suspended.

B) She wants the audience to inspect the thread's knots as the main subject.

C) She intends to sew the button back onto the new cloth immediately.

D) She expects the cardboard cloud to turn the brass into a different material.

163. Which feature most clearly makes the narration an example of personification?

A) The coat is changed to suit a different cloth.

B) A lamp makes the brass surface shine.

C) The theater uses paper for both sky and sea.

D) The button thinks about its usefulness and watches the ship with interest.

164. What does the final contrast between keeping something closed and waiting for a story to open emphasize?

A) The button has secretly returned to the coat before the play starts.

B) The theater cannot operate unless its curtains remain permanently closed.

C) The narrator's role has shifted from practical fastening to imaginative participation.

D) The button's metal has physically changed into paper during the scene.

Informational Passage: Two Shapes in a Soap Film

A soap film stretched across a flat wire loop can look nearly flat, while a free-floating soap bubble is usually round. The contrast does not mean that the liquid follows two unrelated principles. Its shape depends on both a common tendency and the restrictions placed on it.
Surface tension makes a liquid surface behave somewhat like a stretched skin. Reducing surface area reduces the energy associated with that surface. A film therefore tends toward a smaller area, but it cannot ignore the boundaries or enclosed space involved in a particular situation.
On a flat wire loop, the rim fixes the film's edge. A nearly flat sheet can span the opening bounded by that rim with little area. A bubble, however, encloses air. In an ideal comparison between shapes enclosing the same volume, a sphere uses the least surface area. The round shape follows from minimizing area while preserving an enclosed volume, not from a rule requiring every film to become a sphere.

Real shapes can be affected by air currents, gravity, or contact with other bubbles. Such complications do not make the simple comparison useless. It identifies which restriction matters: the fixed rim for the flat film and the enclosed volume for the bubble. Asking only whether the material is soapy misses the conditions that help determine its form.

165. What fixes the edge of the nearly flat film described in the passage?

A) the volume of air inside a closed sphere

B) the wire loop's rim

C) the direction of every nearby air current

D) the distance between two separate bubbles

166. Which general explanation accounts for both the flat film and the round bubble?

A) The same tendency to reduce surface area can produce different shapes under different restrictions.

B) A soap film must always form a sphere regardless of its boundary.

C) The flat film and the bubble obey completely unrelated principles.

D) Real air currents make every simplified explanation of shape worthless.

167. A flat film spans a rectangular wire loop. Why would this example NOT disprove the explanation of a spherical free-floating bubble?

A) The explanation says surface tension exists only in perfectly circular wire loops.

B) The loop fixes a boundary, while the bubble comparison concerns shapes enclosing the same volume.

C) A rectangular loop automatically removes all surface energy from its liquid.

D) A bubble's sphere is explained by the complete absence of enclosed air.

168. In the phrase "span the opening," span most nearly means ____.

A) measure the age of

B) stretch across

C) break into many pieces inside

D) move away from

169. How is the explanation organized after its opening comparison?

A) It lists the dates on which different soap formulas were invented.

B) It gives a personal account of buying wire and then abandons the shape question.

C) It argues that the two shapes must be identical and ends by denying any restrictions exist.

D) It presents a shared physical tendency, applies it under two restrictions, and then notes complications in real examples.

Humanities Passage: What a Promise Changes

A prediction and a promise can both concern the future, but they do different work. Someone who predicts a call at noon says what is expected to happen. Someone who promises, "I will call you at noon," undertakes a responsibility. The second speaker has done more than describe a likely event.

That responsibility can change another person's plans. A listener may remain available, postpone an errand, or prepare information for the conversation. The promise creates a basis for reliance: another person can organize an action around the speaker's commitment.

Making a promise does not create complete control over circumstances. An unexpected broken connection could prevent a sincere caller from getting through. That possibility does not turn the original commitment into an empty prediction. The speaker still has reason to make a reasonable effort and, when possible, explain a failure rather than treat the listener's waiting as irrelevant.

The difference also explains why a promise need not be formal or dramatic. A few ordinary words can establish

an expectation when both people understand them as a commitment. Their importance lies partly in what someone else may do because of them. To understand a promise, then, consider not only the future action named by the speaker but also the practical position of the person who is being asked to count on it.

170. In the author's example, what action does the promising speaker undertake?

A) postponing the listener's errand

B) calling the listener at noon

C) repairing every possible broken connection

D) predicting the exact length of the conversation

171. Which explanation of a promise brings together the passage's points about the speaker's commitment and the listener's plans?

A) It guarantees control over every circumstance that could affect a future action.

B) It matters only when expressed in a formal ceremony.

C) It removes any reason to explain a failure once an obstacle appears.

D) It creates a responsibility and a basis for another person's planning, rather than merely predicting an event.

172. Why can an unexpected broken connection prevent a call without making the original promise merely a prediction?

A) A sincere promise gives the speaker unlimited control over communication systems.

B) An obstacle automatically cancels every effect on the listener's plans.

C) A promise counts only after the named action has already been completed.

D) A commitment establishes responsibility even when outside circumstances prevent the intended result.

173. As used in "a basis for reliance," reliance most nearly means ____.

A) certainty about every outside event

B) indifference toward a future action

C) dependence on a commitment

D) ownership of a communication device

174. Why does the conclusion direct attention to the person who is asked to count on the promise?

A) It completes the explanation by showing the commitment's consequences from the listener's side.

B) It introduces a claim that listeners must never make plans.

C) It replaces the discussion with a technical explanation of telephone construction.

D) It denies that the speaker's future action has any relevance to the promise.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Which expression writes 0.00072 in scientific notation?

A) 7.2 × 10□

B) 7.2 × 10□³

C) 7.2 × 10□□

D) 72 × 10□□

176. Two different rational numbers are averaged. Which statement must be true about their average?

A) It is always an integer.

B) It is irrational and larger than both.

C) It is equal to one of the original numbers.

D) It is rational and lies strictly between them.

177. How much greater is (−3)□ than −3□?

A) 0

B) 81

C) 162

D) −162

178. Which description applies to the decimal form of every irrational number?

A) It neither terminates nor repeats a fixed block forever.

B) It always ends after finitely many digits.

C) It repeats a fixed block after some point.

D) It can always be written as a ratio of two integers.

179. Which inequality is equivalent to −4x > 20?

A) x > −5

B) x < 5

C) x > 5

D) x < −5

180. The identity a(x + 2) + b(x − 1) = 7x + 5 is true for every real x. What are a and b?

A) a = 3; b = 4

B) a = 4; b = 3

C) a = 5; b = 2

D) a = 2; b = 5

181. Which equation has exactly the two solutions x = −2 and x = 4?

A) |x − 1| = 3

B) |x + 1| = 3

C) |x − 1| = 2

D) |x − 3| = 1

182. Let $f(x) = x^2 - x$. For what value of x is $f(x) = f(x + 1)$?

A) 0

B) −1

C) 1

D) 2

183. Two concentric circles have radii 7 and 4 units. What is the area of the ring between them?

A) 9π square units

B) 65π square units

C) 11π square units

D) 33π square units

184. A cube net has four squares in a horizontal row, labeled P, Q, R, S from left to right. Square T is attached above Q, and U below Q. When the net is folded into a cube, which face is opposite P?

A) Q

B) R

C) S

D) T

185. In triangle ABC, angle B is 50° and angle C is 70°. The internal bisectors of B and C meet at I. What is angle BIC?

A) 60°

B) 100°

C) 120°

D) 150°

186. A measuring scale reads 20 when the true value is 0, and 120 when the true value is 50. The scale changes uniformly between these points. What true value corresponds to a reading of 70?

A) 35

B) 25

C) 50

D) 70

187. Which listed length requires the most digits after the decimal point when expressed exactly in meters, without unnecessary trailing zeros?

A) 1 millimeter

B) 1 centimeter

C) 1 meter

D) 1 kilometer

188. A bar chart shows values 100 and 110, but its vertical axis begins at 90. The visible bar heights are therefore 10 and 20. Why would using those visible heights as a ratio exaggerate the difference?

A) They suggest equal values, while the actual values have a ratio of 2 to 1.

B) They show the true values directly because subtracting 90 preserves every ratio.

C) They prove that the value 110 is 90% smaller than the value 100.

D) They suggest a ratio of 2 to 1, while the actual values have a ratio of 1.1 to 1.

189. One of two envelopes is chosen with probability 1/2 each. Then one token is chosen uniformly from that envelope. Envelope A contains 1 red and 1 blue token; B contains 3 red and 1 blue token. What is the probability of red?

A) 2/3

B) 5/8

C) 3/8

D) 1/2

190. Across several days, higher outdoor-concert attendance is associated with higher ice-cream sales. Which conclusion is justified by this association alone?

A) The concerts necessarily cause every increase in sales.

B) Buying ice cream necessarily causes people to attend concerts.

C) The two recorded quantities tend to be higher together.

D) No other factor could affect either quantity.

191. Which numeral represents three million four thousand?

A) 3,040,000

B) 3,004,000

C) 3,400,000

D) 3,000,004

192. What positive number has a square equal to 0.0081?

A) 0.9

B) 0.009

C) 0.09

D) 0.81

193. Let r be a nonzero irrational number. Which expression must be rational?

A) r/r

B) r + 1

C) 2r

D) r^2

194. Which expression is equal to $x^2 - 9$ for every real x?

A) $(x - 3)^2$

B) $(x - 3)(x + 3)$

C) $x(x - 9)$

D) $(x - 9)(x + 9)$

195. Perpendicular nonvertical lines have slopes whose product is −1. Which equation describes a line perpendicular to a line of slope 2 and passing through (4, 1)?

A) $y = 2x - 7$

B) $y = -2x + 9$

C) $y = -x/2 + 3$

D) $y = x/2 - 1$

196. Which equation represents a direct proportional relationship between x and y?

A) y = 7x

B) y = x + 7

C) xy = 7

D) y = 7/x + 1

197. What real value of x satisfies |x| + x = 8?

A) −4

B) 0

C) 8

D) 4

198. The midpoints of adjacent sides of an 8-by-14 rectangle are joined to form an inner quadrilateral. What is the inner quadrilateral's area?

A) 28 square units

B) 84 square units

C) 112 square units

D) 56 square units

199. A rectangular box measures 3 by 4 by 12 units. The diagonal across its 3-by-4 base and a vertical 12-unit edge form a right triangle. What is the straight-line distance between opposite vertices of the box?

A) 5 units

B) 17 units

C) 19 units

D) 13 units

200. What is the smallest positive rotation about the center that maps a regular pentagon onto itself?

A) 36°

B) 72°

C) 108°

D) 144°

201. A strip will be sewn exactly once around the entire edge of a tent flap, with no overlap. Which measure determines the needed strip length?

A) area

B) volume

C) perimeter

D) mass

202. Two pointers begin together at the top of a circular dial. They rotate clockwise at constant rates of 6° per second and 2° per second. After how many seconds do they first point in the same direction again?

A) 90 seconds

B) 45 seconds

C) 60 seconds

D) 180 seconds

203. A board will be cut into eight pieces, each exactly 25 centimeters long, with no trimming at either end. Each saw cut removes 2 millimeters of wood. What initial board length is required?

A) 200 centimeters

B) 201.4 centimeters

C) 201.6 centimeters

D) 214 centimeters

204. A game awards 8 points with probability 1/4 and 0 points with probability 3/4. Its expected score is the sum of each score multiplied by its probability. What is the expected score?

A) 2 points

B) 3 points

C) 6 points

D) 8 points

205. One number is added to the data set 2, 3, 3, 8. The new five-number set has equal mean, median, and unique mode. What number was added?

A) 0

B) 3

C) 5

D) −1

206. Three tiles marked A, B, and C are placed in a random left-to-right order, with all six orders equally likely. What is the probability that A is immediately to the left of B?

A) 1/6

B) 1/2

C) 1/3

D) 2/3

207. What is the sum 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128?

A) 254

B) 256

C) 511

D) 255

208. What is the value of 1 − 2 + 3 − 4 + ... + 99 − 100?

A) −50

B) −100

C) 0

D) 50

209. A positive integer has an odd number of positive factors. What must be true about the integer?

A) It is a prime number.

B) It is an even number.

C) It is a perfect square.

D) It is divisible by 4.

210. For nonzero real numbers a, b, and c, which expression always equals a ÷ (b ÷ c)?

A) ac/b

B) a/(bc)

C) ab/c

D) bc/a

211. A set of ordered pairs represents a function if each input has exactly one output. Which set represents a function?

A) {(1, 2), (1, 3), (2, 4)}

B) {(0, 1), (0, 2), (3, 2)}

C) {(1, 4), (2, 4), (3, 4)}

D) {(−1, 0), (2, 3), (2, 5)}

212. Which description gives all real numbers satisfying x − 2 > 3 OR x + 4 < 1?

A) x > 5 or x < −3

B) −3 < x < 5

C) x > −3

D) x < 5

213. Squaring both sides of $\sqrt{(x + 5)} = x - 1$ produces the candidates x = −1 and x = 4. Which candidates actually satisfy the original equation?

A) Only x = −1

B) Both candidates

C) Only x = 4

D) No real candidate

214. A function satisfies f(f(x)) = x for every real x: applying the same function twice returns the original input. If f(6) = 11, what is f(11)?

A) 11

B) 6

C) 17

D) 22

215. A semicircle has radius 6 units. What is its complete perimeter, including the straight side?

A) 6π units

B) 12π units

C) 6π + 12 units

D) 3π + 12 units

216. Which shape can have exactly one pair of parallel sides?

A) square

B) rectangle

C) parallelogram

D) trapezoid

217. A dilation has center (1, 2) and scale factor 3. Multiply each displacement from the center by 3, then add the center back. Where does P(3, −1) move?

A) (9, −3)

B) (7, −7)

C) (6, −9)

D) (7, −1)

218. All pigment percentages are by volume. An artist mixes 100 milliliters of ink containing 20% pigment with ink containing 50% pigment. Assuming volumes add, how much of the 50% ink makes a mixture containing 30% pigment?

A) 20 milliliters

B) 30 milliliters

C) 100 milliliters

D) 50 milliliters

219. In the Gregorian calendar, a century year is a leap year only if it is divisible by 400. Which listed century year contains February 29?

A) 2000

B) 2100

C) 2200

D) 2300

220. For data set A, the first and third quartiles are 10 and 14. For B, they are 8 and 17. The middle-half span is third quartile minus first quartile. Which statement is correct?

A) A has span 9, and B has span 4.

B) A has span 24, and B has span 25.

C) A has span 14, and B has span 17.

D) A has span 4, and B has span 9.

221. A device displays A, B, or C on each of three independent presses, with equal probabilities. What is the probability that at least two of the three displayed symbols are the same?

A) 2/3

B) 7/9

C) 1/3

D) 2/9

222. A stock record lists 5 small jackets, k medium jackets, and 7 large jackets. If medium is the unique mode of jacket size, what is the least possible integer k?

A) 7

B) 8

C) 12

D) 2

223. A prime number has exactly two positive factors; a composite number has more than two. How is 1 classified?

A) Prime, because it is odd

B) Neither prime nor composite, because it has exactly one positive factor

C) Composite, because it is a whole number

D) Prime, because it is a factor of itself

224. A negative exponent means take the reciprocal of the corresponding positive power. What is (3/2) raised to the power −2?

A) 9/4

B) −9/4

C) −4/9

D) 4/9

225. Which fraction, in lowest terms, lies strictly between 2/5 and 3/7 and has the smallest possible denominator?

A) 3/8

B) 4/9

C) 7/17

D) 5/12

226. A shipping calculator takes a package's mass as input and returns the postage for that mass. Which quantity is the dependent variable?

A) the mass entered

B) the postage returned

C) the name of the mass unit

D) the package's identification code

227. Which equation describes the vertical line passing through (−2, 5)?

A) $y = 5$

B) $x = -2$

C) $y = -2$

D) $x = 5$

228. Positive quantities satisfy $y = kx^2/z$, where k is a fixed positive constant. If x is tripled and z is multiplied by 6, the new y is what multiple of the original y?

A) 3/2

B) 1/2

C) 54

D) 9

229. For $y = (x - h)^2 + k$, the vertex is (h, k). The graph of $y = x^2$ moves 3 units right and 2 units down. Which equation describes the moved graph?

A) $y = (x + 3)^2 - 2$

B) $y = (x - 3)^2 + 2$

C) $y = (x + 3)^2 + 2$

D) $y = (x - 3)^2 - 2$

230. In a plane, what is the set of all points exactly 4 units from a fixed point O?

A) a circle centered at O

B) a straight line through O

C) a filled disk centered at O

D) a square with center O

231. A semicircular sheet of radius 10 units is rolled into a cone without stretching or overlap. Its curved edge becomes the circle around the cone's base. What is the base radius?

A) 10 units

B) 20 units

C) 5 units

D) 10π units

232. A cube has edge length 2 units. A shortest surface path between opposite vertices becomes a diagonal when two adjacent faces are unfolded into a rectangle. What is the path length?

A) 2√5 units

B) 2√3 units

C) 6 units

D) 4 units

233. A recording timer displays 02:07.50, meaning minutes:seconds.hundredths of a second. What duration does it show in seconds?

A) 2.075 seconds

B) 27.5 seconds

C) 127.5 seconds

D) 127.05 seconds

234. Prototype cart A uses 8 liters per 100 kilometers. Cart B travels 15 kilometers per liter. For a 300-kilometer trip at those constant rates, which cart uses less fuel, and by how much?

A) Cart A uses 4 liters less.

B) Cart B uses 9 liters less.

C) Cart A uses 7 liters less.

D) Cart B uses 4 liters less.

235. A scale can err by at most 2 grams in either direction on each reading. A filled container reads 164 grams; the empty container reads 48 grams. What is the narrowest range guaranteed to include every possible actual net mass of the contents?

A) 112 through 120 grams

B) 114 through 118 grams

C) 116 through 120 grams

D) 110 through 122 grams

236. The mean absolute deviation is the average of the absolute distances of the values from their mean. What is the mean absolute deviation of 8, 11, and 17?

A) 12

B) 10

C) 10/3

D) 3

237. A wait-time histogram represents frequency by bar area. The interval 0–2 minutes has height 6; the interval 2–5 minutes has height 5. Which interval contains more observations, and by how many?

A) 0–2 minutes; 1 more

B) 2–5 minutes; 3 more

C) 2–5 minutes; 1 more

D) 0–2 minutes; 3 more

238. Four distinct cards numbered 1 through 4 are randomly assigned, one each, to four places numbered 1 through 4. Every assignment is equally likely. What is the probability that exactly three cards match their place numbers?

A) 1/4

B) 1/6

C) 0

D) 1/24

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which choice correctly contracts the words we will?

A) well

B) wel'l

C) we'll

D) w'ell

240. In a formal American business letter, which salutation uses the conventional ending punctuation?

A) Dear Director Patel;

B) Dear Director Patel.

C) Dear Director Patel?

D) Dear Director Patel:

241. Two brothers-in-law each have a workbench. Which phrase correctly refers to both men's workbenches?

A) the brothers-in-law's workbenches

B) the brother-in-law's workbenches

C) the brothers'-in-law workbenches

D) the brother-in-laws' workbenches

242. Which sentence correctly capitalizes a brand name while leaving the general product and landscape words lowercase?

A) We used a nikon camera to photograph the dunes.

B) We used a Nikon camera to photograph the dunes.

C) We used a Nikon Camera to photograph the dunes.

D) We used a Nikon camera to photograph the Dunes.

243. Which sentence distinguishes the formal numbered course title from the general subject?

A) She enrolled in oceanography 205 after taking introductory Oceanography.

B) She enrolled in Oceanography 205 after taking introductory Oceanography.

C) She enrolled in oceanography 205 after taking introductory oceanography.

D) She enrolled in Oceanography 205 after taking introductory oceanography.

244. Select the standard spelling of the noun meaning a difficult choice between alternatives.

A) dilemna

B) dilemma

C) dilema

D) dillemma

245. Which spelling correctly completes the sentence?
The diagram showed the company's management ____.

A) heirarchy

B) hiearchy

C) hierachy

D) hierarchy

246. Which word correctly completes the sentence?
The telescope, ____ lens had cracked, was placed in its case.

A) who's

B) whom

C) whose

D) which

247. Which form correctly completes the sentence?
The mechanic had the axle ____ before the next trip.

A) repair

B) repairing

C) repaired

D) to repair

248. Which phrase correctly completes the sentence in formal written English?
No sooner ____ than the alarm rang.

A) had the door closed

B) the door had closed

C) the door has closed

D) did the door closed

249. Which pair correctly completes the sentence?
Please ____ the guest that the path is safe, and ____ that the gate is closed.

A) ensure; assure

B) assure; ensure

C) assure; assure

D) ensure; ensure

250. Which pair correctly completes the sentence?
The inventor devised an ____ clasp; the child answered with an ____ sincerity that showed no attempt to deceive.

A) ingenuous; ingenious

B) ingenious; ingenious

C) ingenious; ingenuous

D) ingenuous; ingenuous

Editing Passage: Two Ways to Bind a Notebook

(1) A notebook can be bound with thread or with adhesive. (2) Thread binding joins folded groups of pages with stitches. (3) This construction can allow the notebook to lie relatively flat when open. (4) It often involves more assembly steps and can take longer to produce. (5) Adhesive binding joins a stack of pages by gluing their prepared edges into the cover. (6) This method can be efficient when many notebooks must be produced. (7) Some adhesive-bound notebooks resist lying flat, although opening behavior varies with their materials and

construction. (8) A buyer choosing between the methods should consider how the notebook will be used rather than assuming that one method is best for every purpose.

251. The writer wants to reorganize sentences 2–7 by feature: first construction, then opening behavior, then production. Which order of sentence pairs accomplishes that goal?

A) (2, 3), then (4, 5), then (6, 7)

B) (3, 7), then (2, 5), then (4, 6)

C) (4, 6), then (3, 7), then (2, 5)

D) (2, 5), then (3, 7), then (4, 6)

252. Which sentence best combines the information in sentences 2 and 5 while emphasizing the difference between the methods?

A) Thread binding stitches folded page groups together, whereas adhesive binding glues prepared page edges into the cover.

B) Both methods stitch folded page groups together before gluing the same stitches into the cover.

C) Adhesive binding stitches folded page groups, whereas thread binding glues their edges into the cover.

D) Thread binding and adhesive binding are two names for a process whose construction is identical.

253. Which addition immediately after sentence 1 would best preview the comparison for readers unfamiliar with bookbinding?

A) Both methods attach pages to a cover, but they differ in construction, opening behavior, and production.

B) Only experts should purchase notebooks, because ordinary readers cannot understand their covers.

C) Thread binding must be best in every situation because this passage discusses it first.

D) Before buying a notebook, readers should learn the complete history of every writing instrument.

254. The original sentence is "We waited through the cold afternoon before the gate opened." Which excerpt uses an ellipsis to mark the omitted words?

A) "We waited, before the gate opened."

B) "We waited; before the gate opened."

C) "We waited: before the gate opened."

D) "We waited ... before the gate opened."

255. Several pilots attend a briefing. The name Lena Cho is essential to identify which pilot will speak first. Which sentence punctuates that identifying name accordingly?

A) The pilot Lena Cho will speak first.

B) The pilot, Lena Cho, will speak first.

C) The pilot Lena Cho, will speak first.

D) The pilot, Lena Cho will speak first.

256. Which sentence correctly capitalizes the name of a historical era?

A) Artists of the renaissance studied proportion.

B) Artists of the Renaissance studied Proportion.

C) Artists of the Renaissance studied proportion.

D) Artists of the Renaissance Studied proportion.

257. When diesel names an engine type rather than a person, which sentence uses standard capitalization?

A) The harbor tug uses a Diesel engine.

B) The harbor tug uses a diesel Engine.

C) The Harbor Tug uses a diesel engine.

D) The harbor tug uses a diesel engine.

258. Which spelling correctly completes the sentence?
The cleaner connected the hose to the ____.

A) vaccum

B) vacum

C) vacuum

D) vaccuum

259. Select the correctly spelled word for a meeting arranged in advance.

A) rendevous

B) rendezvouz

C) rendezvousz

D) rendezvous

260. Which word is spelled correctly?

A) liason

B) liaison

C) laison

D) liasion

261. Which sentence uses herself to emphasize the subject rather than as the receiver of the verb's action?

A) The potter photographed herself.

B) The potter accidentally cut herself.

C) The potter congratulated herself.

D) The potter herself signed the base.

262. Which pair of articles correctly completes the sentence?
____ honest witness gave ____ useful account.

A) A; a

B) An; a

C) An; an

D) A; an

263. Which pair correctly completes the sentence?
He used to ____ at noon; now he is used to ____ at dawn.

A) swim; swimming

B) swimming; swim

C) swim; swim

D) swimming; swimming

264. Choose the pair that means accept officially and modify to fit, respectively.
The board will ____ the proposal, and the technicians will ____ the old frame.

A) adopt; adapt

B) adapt; adopt

C) adopt; adopt

D) adapt; adapt

265. Which pair correctly completes the sentence?
An introduction will ____ the demonstration; afterward, the speaker will ____ to questions.

A) precede; proceed

B) proceed; precede

C) precede; precede

D) proceed; proceed

Editing Passage: A Corrected Demonstration Notice

(1) The cloth-dyeing demonstration was originally announced for October 14 in the South Room at 4 p.m. (2) The organizer has moved it to October 21 and the North Room; the starting time will remain 4 p.m. (3) Anyone whose place has already been confirmed will keep that place without taking further action. (4) People without a reservation may still request one before October 18. (5) Originally printed tickets will be accepted despite showing the old date and room. (6) This notice corrects the earlier announcement; it does not announce a second demonstration. (7) Questions about access to the new room may be directed to the organizer.

266. Which heading would best draw attention to both actual changes announced in the notice?

A) Second Demonstration Added for October 14

B) Starting Time Changed and All Reservations Canceled

C) Demonstration Moved to October 21 in the North Room

D) South Room Event Continues at a New Starting Time

267. Which sentence should NOT be added to the notice because it would mislead people who already have confirmed places?

A) Your confirmed place remains reserved for the rescheduled demonstration.

B) Please cancel your old reservations and book again before October 18.

C) You may use your originally printed ticket even though its date is outdated.

D) Please note the new date and room before attending.

268. On October 16, someone without a reservation asks for the relevant information. Which short reply accurately applies the notice without promising a place?

A) Your place is already guaranteed for October 21 in the North Room at 4 p.m.

B) You may request a place before October 18 for the October 21 demonstration in the North Room at 4 p.m.

C) Request a place before October 18 for the October 14 event in the South Room at 4 p.m.

D) You may wait until after October 21 to request a place for that day's demonstration.

269. Which choice uses standard American comma grouping for the whole number 1234567?

A) 12,34,567

B) 123,45,67

C) 1,23,45,67

D) 1,234,567

270. The original quotation is "She returned before sunset." A writer adds Dr. Miro to explain who she means. Which version uses the standard editorial marks for an added clarification?

A) "She {Dr. Miro} returned before sunset."

B) "She /Dr. Miro/ returned before sunset."

C) "She [Dr. Miro] returned before sunset."

D) "She ... Dr. Miro ... returned before sunset."

271. Which sentence correctly uses a colon to introduce an explanation of the complete statement before it?

A) The casting had one serious flaw: a crack ran through its base.

B) The casting had one serious flaw, a crack ran through its base.

C) The casting had: one serious flaw a crack ran through its base.

D) The casting had one: serious flaw a crack ran through its base.

272. In a two-word scientific species name, capitalize the first word but not the second. Which choice follows this rule?

A) Canis lupus

B) canis lupus

C) Canis Lupus

D) CANIS LUPUS

273. Which sentence uses conventional capitalization for the doctoral-degree abbreviation shown?

A) She holds a ph.d. in geology.

B) She holds a PH.d. in geology.

C) She holds a pH.D. in geology.

D) She holds a Ph.D. in geology.

274. Which spelling correctly names a burial ground?

A) cemetary

B) cemeterry

C) cemetery

D) cemetry

275. Which spelling correctly completes the sentence?
The recipe calls for steamed ____.

A) brocolli

B) broccoli

C) brocoli

D) broccolli

276. Choose the standard negative question tag that asks for confirmation.
The gates have closed, ____?

A) haven't it

B) haven't they

C) hasn't they

D) have they

277. Which word correctly completes the sentence?
____ seemed that the gate was open.

A) It

B) There

C) They

D) These

278. In which sentence can that be deleted while keeping the same meaning and a grammatically complete sentence?

A) The rope that we bought is strong.

B) The rope that supports the awning is strong.

C) The rope that crosses the channel is strong.

D) The rope that protects the opening is strong.

279. Which pair correctly completes the sentence?
Despite ____ weather, the crew was ____ to postponing the crossing.

A) averse; adverse

B) adverse; adverse

C) adverse; averse

D) averse; averse

280. Choose the pair that means careful not to attract attention and separate, respectively.
A ____ inquiry concerned four ____ sections of the display.

A) discrete; discreet

B) discreet; discrete

C) discreet; discreet

D) discrete; discrete

281. A report compares a tunnel and a viaduct, in that order. In a later sentence, the former refers to ____ and the latter refers to ____.

A) the viaduct; the tunnel

B) the tunnel; the tunnel

C) the viaduct; the viaduct

D) the tunnel; the viaduct

Editing Passage: Describing What an Image Contributes

(1) A description accompanying an image should communicate the information that readers need from that image. (2) The same photograph may need different wording in different settings. (3) In an article about stoneworking techniques, an image of a mason holding a chisel should help readers understand the tool's position and action. (4) In a biographical profile, the same image might instead identify the mason and the place of work. (5) Descriptions should not automatically list every visible detail, such as shoelaces or distant clouds. (6) If a caption already gives an important fact, the description should add what the image contributes rather than merely repeat that fact. (7) The guiding question is what purpose that particular image serves for its readers.

282. Which additional example after sentence 4 would best extend the principle that a description depends on the image's purpose?

A) A cookbook and a chef's profile should use identical descriptions whenever their pages contain the same photograph.

B) A photograph with more visible objects always needs a longer description, whatever topic the surrounding text discusses.

C) A cookbook might describe how dough is folded, while a chef's profile might identify the person in the same photograph.

D) Descriptions should list distant background objects before explaining anything connected with the article's topic.

283. An article explains chisel placement. Its caption already says "Mason Lea Ortiz at work." The photograph shows a chisel resting on the stone's upper edge, a raised hammer, and clouds in the distance. Which description best applies sentences 3, 5, and 6?

A) Mason Lea Ortiz at work.

B) The chisel rests on the stone's upper edge, with the mason holding the hammer above it.

C) Several clouds appear beyond the workshop in the distance.

D) Lea Ortiz is the finest stoneworker in the entire region.

284. Which mark should end this ordinary declarative sentence?
The guard asked whether the gate was locked____

A) ?

B) ,

C) ;

D) .

285. Use double quotation marks for Rina's speech and single quotation marks for the sign's exact words. Which sentence is punctuated correctly in American English?

A) Rina said, "The sign reads 'Keep clear'".

B) Rina said "The sign reads 'Keep clear.'"

C) Rina said, "The sign reads 'Keep clear".'

D) Rina said, "The sign reads 'Keep clear.'"

286. A witness addresses a judge by a formal title. Which sentence uses standard capitalization?

A) The diagram is ready, your honor.

B) The diagram is ready, Your honor.

C) The diagram is ready, Your Honor.

D) The diagram is ready, your Honor.

287. A reviewer uses the exact phrase a quiet farewell within a larger sentence, not as a complete quoted statement or a title. Which capitalization is correct?

A) The reviewer called the final movement "A quiet farewell."

B) The reviewer called the final movement "A Quiet Farewell."

C) The reviewer called the Final Movement "a quiet farewell."

D) The reviewer called the final movement "a quiet farewell."

288. Which spelling correctly names a fine ceramic material?

A) porcelin

B) porcelainn

C) porcelain

D) porcelane

289. Which spelling correctly completes the sentence?
Sudden ____ disturbed the smooth airflow.

A) turbulance

B) turbelence

C) turbulencee

D) turbulence

290. Choose the correctly spelled verb meaning replace something with a newer alternative.

A) supercede

B) supersede

C) superceed

D) superseed

291. Which pair correctly completes the sentence?
The instructions were ____, and the readers became ____.

A) confused; confusing

B) confusing; confusing

C) confusing; confused

D) confused; confused

292. Each diver checked the other diver's clip. Which sentence preserves that mutual meaning with the correct possessive pronoun phrase?

A) The divers checked their own clips.

B) The divers checked each other's clips.

C) The divers checked each others' clips.

D) The divers checked each other clips.

293. The opening is scheduled for tomorrow. Which verb correctly completes the future-time clause?
We will begin when the director ____.

A) will arrive

B) arrived

C) arrives

D) had arrived

294. Which sentence uses flaunt or flout correctly?

A) The visitor was warned not to flout the rules.

B) She flouted her new bracelet for everyone to admire.

C) He flaunted the ban by entering a closed area.

D) The tailor flouted the vibrant fabric in the display window.

295. Which word correctly completes the sentence?
The counselor asks open questions to ____ responses from quiet participants.

A) elicit

B) illicit

C) emit

D) omit

296. Which word correctly completes the sentence?
The narrow table stands directly ____ the armchair.

A) beside

B) besides

C) between

D) through

Editing Passage: A Word with a Second View

(1) A physical window is an opening that frames a view of part of the surroundings. (2) Its edges select what can be seen from a particular position. (3) A computer window is a bounded region of a screen containing a document or a program's contents. (4) The display outside that region may contain other information, just as a physical window shows only part of a larger scene. (5) The shared name draws attention to a limited resemblance: both uses involve a framed view. (6) The comparison does not claim that the two things share every material or mechanism. (7) A familiar word can thus help a reader understand something new while leaving important differences intact.

297. Sentences 1–2 form the first paragraph. Which topic sentence would best introduce a second paragraph containing sentences 3–7?

A) Every computer display must be constructed exactly like an opening in a wall.

B) A familiar name can acquire a new use through one important resemblance.

C) Physical windows have appeared in many buildings of different sizes.

D) The history of every word can be explained by the invention of computer screens.

298. A writer proposes adding: "Every physical feature of the original kind must also exist in the newer kind." How should this proposal be handled?

A) Omit it because it turns a limited resemblance into an unsupported claim about all features.

B) Put it before sentence 1 as a definition of every possible opening.

C) Put it after sentence 4 as proof that screen regions must contain glass.

D) Put it after sentence 7 as a conclusion that eliminates all differences.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. B	2. C	3. A	4. B	5. C	6. D	7. A	8. C
9. B	10. A	11. B	12. D	13. D	14. C	15. A	16. D
17. B	18. D	19. A	20. C	21. C	22. A	23. D	24. D
25. A	26. B	27. C	28. B	29. C	30. B	31. A	32. C
33. D	34. A	35. D	36. C	37. C	38. B	39. C	40. D
41. B	42. A	43. B	44. A	45. D	46. D	47. C	48. D
49. A	50. A	51. C	52. B	53. A	54. D	55. B	56. B
57. C	58. D	59. C	60. B				

Quantitative Skills

61. B	62. A	63. C	64. B	65. C	66. A	67. D	68. C
69. B	70. D	71. A	72. B	73. D	74. D	75. B	76. C
77. D	78. A	79. C	80. B	81. A	82. D	83. C	84. A
85. B	86. C	87. D	88. A	89. B	90. D	91. C	92. A
93. B	94. C	95. B	96. D	97. C	98. A	99. D	100. C
101. A	102. B	103. D	104. A	105. C	106. B	107. A	108. B
109. C	110. D	111. A	112. D				

Reading Comprehension

113. C	114. B	115. D	116. A	117. C	118. D	119. A	120. D
121. C	122. A	123. C	124. A	125. B	126. B	127. B	128. D
129. B	130. D	131. B	132. C	133. A	134. C	135. B	136. C
137. D	138. A	139. D	140. D	141. A	142. C	143. A	144. B
145. D	146. B	147. A	148. B	149. D	150. D	151. A	152. C
153. C	154. B	155. C	156. D	157. A	158. B	159. A	160. C
161. C	162. A	163. D	164. C	165. B	166. A	167. B	168. B
169. D	170. B	171. D	172. D	173. C	174. A		

Mathematics

175. C	176. D	177. C	178. A	179. D	180. B	181. A	182. A
183. D	184. B	185. C	186. B	187. A	188. D	189. B	190. C
191. B	192. C	193. A	194. B	195. C	196. A	197. D	198. D
199. D	200. B	201. C	202. A	203. B	204. A	205. D	206. C

207. D	208. A	209. C	210. A	211. C	212. A	213. C	214. B
215. C	216. D	217. B	218. D	219. A	220. D	221. B	222. B
223. B	224. D	225. D	226. B	227. B	228. A	229. D	230. A
231. C	232. A	233. C	234. D	235. A	236. C	237. B	238. C

Language

239. C	240. D	241. A	242. B	243. D	244. B	245. D	246. C
247. C	248. A	249. B	250. C	251. D	252. A	253. A	254. D
255. A	256. C	257. D	258. C	259. D	260. B	261. D	262. B
263. A	264. A	265. A	266. C	267. B	268. B	269. D	270. C
271. A	272. A	273. D	274. C	275. B	276. B	277. A	278. A
279. C	280. B	281. D	282. C	283. B	284. D	285. D	286. C
287. D	288. C	289. D	290. B	291. C	292. B	293. C	294. A
295. A	296. A	297. B	298. A				

Detailed Explanations

Verbal Skills

1. HSPT-V12-001 - Answer B: nearness
Proximity describes how near something is to another thing, rather than its height, direction, or total movement.

2. HSPT-V12-002 - Answer C: hasten
To expedite something is to help it happen more quickly, so hasten expresses the intended action.

3. HSPT-V12-003 - Answer A: affectionate longing for the past
Nostalgia combines fond remembrance with longing for an earlier time, rather than concern about the future or interest in something unfamiliar.

4. HSPT-V12-004 - Answer B: convinced
Someone skeptical doubts an explanation, while someone convinced accepts it as true rather than continuing to question it.

5. HSPT-V12-005 - Answer C: halting
A fluent delivery proceeds smoothly, whereas a halting delivery is interrupted by pauses or hesitation that disrupt its flow.

6. HSPT-V12-006 - Answer D: uniform
A heterogeneous mixture contains unlike components, whereas uniform describes a mixture that is consistent throughout rather than varied.

7. HSPT-V12-007 - Answer A: credit
Blame is assigned for a fault, just as credit is assigned for an achievement; both are judgments attributed to their respective grounds.

8. HSPT-V12-008 - Answer C: decompress
Decrypting reverses encryption, just as decompressing reverses compression; the second operation undoes the change made by the first.

9. HSPT-V12-009 - Answer B: home
Exile involves being forced from a country, while eviction involves being forced from a home; each action is paired with the place left.

10. HSPT-V12-010 - Answer A: corridor
A corridor is a constructed passage, like a canal, duct, or tunnel; the other choices name natural landscape features.

11. HSPT-V12-011 - Answer B: intersect
Intersect means cross or meet, while encircle, surround, and envelop describe an all-around relation to something.

12. HSPT-V12-012 - Answer D: series; species
One series and several series use the same spelling, as do one species and several species; the other pairs change in the plural.

13. HSPT-V12-013 - Answer D: The flag is red.
Waiting is prescribed only in the red-flag branch when the bell sounds; a sounding bell alone does not override the nonred instruction.

14. HSPT-V12-014 - Answer C: M, K, L, J
Exchanging the middle flags gives J, L, K, M; reversing that whole row then gives M, K, L, J.

15. HSPT-V12-015 - Answer A: A booth with a seat and a window but no camera feed
The seat is mandatory, while the window and camera feed are alternatives; the seated booth with a window satisfies both parts.

16. HSPT-V12-016 - Answer D: guard
A sentry is a person assigned to stand watch or guard a place, rather than someone merely visiting or receiving service.

17. HSPT-V12-017 - Answer B: unspoken
Tacit approval is understood without being stated directly; the word describes how approval is conveyed, not how long it lasts.

18. HSPT-V12-018 - Answer D: experienced through another person
Vicarious excitement comes through another person's experience, such as sharing a traveler's adventures through that person's account.

19. HSPT-V12-019 - Answer A: central
A peripheral issue lies outside the main concern, while a central issue belongs to the heart of that concern.

20. HSPT-V12-020 - Answer C: unlimited
Finite means having limits or an end, whereas unlimited describes something without the limiting boundary expressed by finite.

21. HSPT-V12-021 - Answer C: malevolent
Benevolent describes wishing others well, while malevolent describes wishing them harm, making the intentions directly opposed.

22. HSPT-V12-022 - Answer A: appearance
A parody humorously exaggerates features of a style, just as a caricature humorously exaggerates features of someone's appearance.

23. HSPT-V12-023 - Answer D: hands
A bilingual person can use two languages; an ambidextrous person can use both hands skillfully, pairing each ability with its domain.

24. HSPT-V12-024 - Answer D: hearing
An aroma is perceived through smell, while musical pitch is perceived through hearing; each quality is paired with its direct sensory channel.

25. HSPT-V12-025 - Answer A: nephew
Nephew names a family relationship, whereas colleague, visitor, and coach identify other kinds of relationships or roles.

26. HSPT-V12-026 - Answer B: smolder
Smolder means burn slowly without a strong flame; gush, trickle, and seep describe different ways liquid moves.

27. HSPT-V12-027 - Answer C: deafening silence
Deafening suggests overwhelming sound, while silence means absence of sound; their literal opposition creates a deliberate contradiction.

28. HSPT-V12-028 - Answer B: Not all visitors choose exactly the same set of postcards.
If everyone chose an identical nonempty set, each postcard in that set would be chosen by everyone, contradicting the stated condition.

29. HSPT-V12-029 - Answer C: M, K, L, N
The Elm files must come first, with urgent M before ordinary K; the Pine files follow, with urgent L before ordinary N.

30. HSPT-V12-030 - Answer B: R
Removing T leaves R and S; adding and removing U leaves that pair again, and removing S leaves R.

31. HSPT-V12-031 - Answer A: restraining line
A tether is a line or similar fastening that restrains an animal by connecting it to a fixed point.

32. HSPT-V12-032 - Answer C: assign to a lower position
To relegate someone is to assign that person a less important position, rather than promote or consult the person.

33. HSPT-V12-033 - Answer D: withdraw it publicly
Recant means take back a previously stated belief or opinion; explaining or delaying discussion does not withdraw the original claim.

34. HSPT-V12-034 - Answer A: divergent
Convergent paths move toward a common point, whereas divergent paths move apart, reversing the direction of their relationship.

35. HSPT-V12-035 - Answer D: unreadable
Legible writing can be read, while unreadable writing cannot; the other choices describe appearance or production without reversing readability.

36. HSPT-V12-036 - Answer C: staggered
Synchronous events occur together in time, whereas staggered events are arranged at different times; the other choices preserve temporal coincidence.

37. HSPT-V12-037 - Answer C: customer
An invitation is addressed to an intended guest; an invoice is addressed to a customer expected to pay for goods or services.

38. HSPT-V12-038 - Answer B: legend
A reference asterisk directs a reader to an explanatory footnote, just as a map symbol directs the reader to its explanation in a legend.

39. HSPT-V12-039 - Answer C: date
Anonymous describes something whose source is unnamed; undated describes something without a stated date, pairing each adjective with missing identifying information.

40. HSPT-V12-040 - Answer D: soon
Soon identifies when something occurs, like yesterday, today, and tomorrow; the other choices indicate location rather than time.

41. HSPT-V12-041 - Answer B: They can honor or remember a person.
Each can express honor or remembrance; none necessarily requests forgiveness, declares disagreement, or commits its speaker to future conduct.

42. HSPT-V12-042 - Answer A: calm, brief, stern
Calmly, briefly, and sternly simply append ly; the other groups require changing y, adding extra letters, or altering the final e.

43. HSPT-V12-043 - Answer B: Flatten it and remove its green tag.
Flattening supplies the missing condition for Station C, and removing one tag satisfies Station S without sacrificing the panel's dryness.

44. HSPT-V12-044 - Answer A: T begins after R starts and before R ends.
R has already started when S begins; T begins after S ends but before R ends, placing T's beginning inside R's interval.

45. HSPT-V12-045 - Answer D: KABYK
KABYK begins and ends with K without adjacent repeated letters; each alternative violates either the endpoint condition or the adjacency condition.

46. HSPT-V12-046 - Answer D: persuade gently
To coax someone is to persuade that person gently or patiently, rather than force compliance through an accusation or command.

47. HSPT-V12-047 - Answer C: driving force
An impetus supplies the force or encouragement that sets something in motion; it is the push behind the change.

48. HSPT-V12-048 - Answer D: made without a consistent reason or rule
An arbitrary decision depends on preference or chance rather than a consistent principle; the alternatives describe systematic grounds for deciding.

49. HSPT-V12-049 - Answer A: compassionate
Callous describes a lack of concern for another person's suffering, while compassionate describes sympathetic concern and sensitivity to that suffering.

50. HSPT-V12-050 - Answer A: authentic
A counterfeit signature falsely imitates a real one; an authentic signature is genuine rather than presented as someone else's work.

51. HSPT-V12-051 - Answer C: freely given consent
Coercion obtains compliance through force or threats; freely given consent reverses the essential lack of voluntary choice.

52. HSPT-V12-052 - Answer B: payment already made
An alibi supports someone's claim of being elsewhere; a receipt supports a claim that a payment occurred, rather than merely being requested.

53. HSPT-V12-053 - Answer A: rotation
Reversal leaves a palindrome's letter order unchanged; rotation leaves a perfect sphere's appearance unchanged, pairing each with a form-preserving transformation.

54. HSPT-V12-054 - Answer D: a basketball game
Intermission temporarily interrupts a play, while halftime temporarily interrupts a basketball game; each pause occurs within an event that will resume.

55. HSPT-V12-055 - Answer B: chilly
Chilly describes a low-temperature sensation; bitter, sour, and salty identify tastes rather than temperature.

56. HSPT-V12-056 - Answer B: perhaps, possibly, presumably
Perhaps, possibly, and presumably qualify certainty; the other groups strengthen certainty, mark consequences, or indicate time.

57. HSPT-V12-057 - Answer C: dust the shelf; dust the pastry with sugar
Dusting a shelf removes dust, while dusting pastry with sugar adds a fine coating; the other verbs keep the same general action.

58. HSPT-V12-058 - Answer D: Does it have shading?
Curves eliminate the pine, leaving crane and boat; asking about shading separates those two possibilities, while the other questions give identical answers for both.

59. HSPT-V12-059 - Answer C: a grandparent
Dara must also be Nora's parent because Nora and Venn share all parents; Orr, as Dara's parent, is therefore Nora's grandparent.

60. HSPT-V12-060 - Answer B: Fenn is directly linked to Ena.
The two-way rule reverses Ena's stated link with Fenn; it neither establishes nor rules out a separate direct link between Ena and Goro.

Quantitative Skills

61. HSPT-Q12-001 - Answer B: 34
After 29, the next tens digit is 3; its ones digit must exceed 3, making 34 the next qualifying integer.

62. HSPT-Q12-002 - Answer A: 16
Seven is prime, so a seven-square rectangle must measure 1 by 7; its perimeter is 2 × (1 + 7) = 16.

63. HSPT-Q12-003 - Answer C: 11
After 12 is removed, the remaining odd lamps are counted in order: skip 1, remove 3, skip 5, remove 7, skip 9, remove 11.

64. HSPT-Q12-004 - Answer B: 10010110
Changing each digit of 01101001 to its opposite gives 10010110, which is the complete block appended at the next stage.

65. HSPT-Q12-005 - Answer C: 888
The first factor doubles while the second halves, so their product remains unchanged rather than doubling or halving.

66. HSPT-Q12-006 - Answer A: 4
Write n = 9k + 4. Then 2n + 5 = 18k + 13 = 9(2k + 1) + 4, giving remainder 4.

67. HSPT-Q12-007 - Answer D: P's output is 5 greater.
P gives w/2 − 5, while Q gives w/2 − 10; subtracting the outputs leaves a difference of 5 regardless of w.

68. HSPT-Q12-008 - Answer C: The two quantities are equal.
The two added values total 2m and therefore have mean m themselves; adding them leaves the overall mean equal to m.

69. HSPT-Q12-009 - Answer B: Quantity B is greater.
Quantity B minus Quantity A is (b − a)/(ab), which is positive because b − a > 0 and ab > 0.

70. HSPT-Q12-010 - Answer D: The relationship cannot be determined from the information given.
The pair 1 and 9 gives product 9, while 5 and 5 gives product 25; both satisfy the stated sum but give opposite comparisons.

71. HSPT-Q12-011 - Answer A: Quantity A is greater.
A full turn contains six 60° triangle angles but only four 90° square angles, so Quantity A is greater.

72. HSPT-Q12-012 - Answer B: Quantity B is greater.
Only the central small cube has no paint, while each of the eight corner cubes has three painted faces; therefore 1 < 8.

73. HSPT-Q12-013 - Answer D: The relationship cannot be determined from the information given.
The four side lengths do not rigidly determine the quadrilateral; the joint between the 2-unit and 3-unit rods can form either an acute or an obtuse angle.

74. HSPT-Q12-014 - Answer D: 1/30
The next difference is 1/5 − 1/6 = (6 − 5)/30 = 1/30, following the stated unit-fraction construction.

75. HSPT-Q12-015 - Answer B: 1, 4, 6, 4, 1
Keep 1 at both ends; the interior entries are 1 + 3 = 4, 3 + 3 = 6, and 3 + 1 = 4.

76. HSPT-Q12-016 - Answer C: 20
Step 7 would give 13 − 7 = 6, but 6 has already appeared; the rule therefore requires 13 + 7 = 20.

77. HSPT-Q12-017 - Answer D: 180
Combining the reciprocals gives (a + b)/(ab) = 30/(ab) = 1/6, so ab = 30 × 6 = 180.

78. HSPT-Q12-018 - Answer A: (−a)/(−b)
The two negative signs cancel in the quotient, leaving a/b; reversing the fraction or changing only one sign does not guarantee equality.

79. HSPT-Q12-019 - Answer C: 33
Within the interval, multiples of 3 leaving remainder 1 on division by 4 are 21 and 33; only 33 leaves remainder 3 on division by 5.

80. HSPT-Q12-020 - Answer B: $n^2 + n + 1$
The product n(n + 1) is even because consecutive integers include an even number; adding 1 makes $n^2 + n + 1$ odd.

81. HSPT-Q12-021 - Answer A: Quantity A is greater.
The final amount is 1.1 × 1.1 = 1.21 times the original, an overall increase of 21%, which exceeds 20%.

82. HSPT-Q12-022 - Answer D: The two quantities are equal.
Any prime dividing xy divides x or y, but cannot also divide x + y without dividing both; therefore no denominator factor cancels.

83. HSPT-Q12-023 - Answer C: The relationship cannot be determined from the information given.
One prime must be 2, but the other is unspecified: 2 + 3 = 5, while 2 + 13 = 15, giving opposite comparisons.

84. HSPT-Q12-024 - Answer A: Quantity A is greater.
The triangle divides the shared perimeter among three equal sides, while the hexagon divides it among six, making each triangle side twice as long.

85. HSPT-Q12-025 - Answer B: Quantity B is greater.
Since angle A exceeds 90° and all three angles total 180°, angles B and C together must total less than 90°.

86. HSPT-Q12-026 - Answer C: The two quantities are equal.
Traveling once around a convex polygon produces one complete turn; the exterior turning angles therefore sum to 360°, regardless of its side count.

87. HSPT-Q12-027 - Answer D: 3
The spellings run eight, five, four, nine, one, seven, six, ten, three, two; therefore three follows ten in this ordering.

88. HSPT-Q12-028 - Answer A: 2, 3, 3
Eight needs 4 + 4; nine needs 4 + 4 + 1; ten needs 4 + 3 + 3. Neither nine nor ten can use two coins.

89. HSPT-Q12-029 - Answer B: 11
After the outer boundary, the inner entries are 6 and 7 across the top, then 11 and 10 below; clockwise reading gives 11 next.

90. HSPT-Q12-030 - Answer D: 1, 1, 2, 1
The first eight runs end at digit 12. The next run lengths are digits 9, 10, and 11: 2, 1, 2, producing 11, 2, 11; the next four digits are 1121.

91. HSPT-Q12-031 - Answer C: 86
The only pair of digits from 1 through 9 with product 48 is 6 and 8; placing the larger digit first gives 86.

92. HSPT-Q12-032 - Answer A: 24
If the middle number is m, the sum is (m − 2) + (14 − m) + (14 − 2) = 24, regardless of m.

93. HSPT-Q12-033 - Answer B: 1/12
The product becomes (1/2) × (2/3) × (3/4) × ... × (11/12); every interior numerator cancels a denominator, leaving 1/12.

94. HSPT-Q12-034 - Answer C: The two quantities are equal.
Multiplying a positive integer by 10 appends exactly one zero to its decimal notation, increasing its digit count by exactly one.

95. HSPT-Q12-035 - Answer B: Quantity B is greater.
A positive product requires matching signs. Their negative sum rules out two positive numbers, so both are negative and $a < 0$.

96. HSPT-Q12-036 - Answer D: The relationship cannot be determined from the information given.
Both $t = 1/4$ and $t = 3/4$ satisfy $t^2 < t$, but the first makes Quantity B greater and the second makes Quantity A greater.

97. HSPT-Q12-037 - Answer C: The two quantities are equal.
Both triangles have the same base and the same perpendicular height, the distance between the parallel lines; their areas therefore agree.

98. HSPT-Q12-038 - Answer A: Quantity A is greater.
The vertex distance is the hypotenuse of a right triangle whose legs are the positive half-width and half-height, so it exceeds either leg.

99. HSPT-Q12-039 - Answer D: The relationship cannot be determined from the information given.
With equal vertical positions, horizontal offsets of 1/4 and 3/4 give overlap areas 3/4 and 1/4, respectively, placing Quantity A on either side of 1/2.

100. HSPT-Q12-040 - Answer C: 1, 0, 0
The next integers are 10, 11, and 12. Only the zero in 10 contributes a loop, so their counts are 1, 0, 0.

101. HSPT-Q12-041 - Answer A: (44, 7)
Dividing 51 by 44 leaves remainder 7, giving (44, 7); dividing 44 by 7 then leaves 2, confirming the following pair.

102. HSPT-Q12-042 - Answer B: 38
The ranges become 33–64, 33–47, 33–39, and 37–39. The middle integer of the final range is 38.

103. HSPT-Q12-043 - Answer D: 44
XL represents 50 − 10 = 40, while IV represents 5 − 1 = 4; together they give 44.

104. HSPT-Q12-044 - Answer A: $5\sqrt{2}$
Since $\sqrt{18} = 3\sqrt{2}$ and $\sqrt{8} = 2\sqrt{2}$, adding the like radical terms gives $5\sqrt{2}$.

105. HSPT-Q12-045 - Answer C: (3, 3)
If the first register is reduced a times, its final value is 9 − 3a. Only 3 fits the choices; moves left, left, right produce (4, 4), (2, 5), (3, 3).

106. HSPT-Q12-046 - Answer B: 5
With a = 0, the triples are (0,0,5), (0,1,4), and (0,2,3). With a = 1, they are (1,1,3) and (1,2,2). Larger a is impossible.

107. HSPT-Q12-047 - Answer A: Quantity A is greater.
Quantity A lies above n, while Quantity B lies below n; therefore Quantity A must exceed Quantity B, even when n is negative.

108. HSPT-Q12-048 - Answer B: Quantity B is greater.
The middle number stays unchanged, so the median increases by 0; the sum increases by 5 across five entries, raising the mean by 1.

109. HSPT-Q12-049 - Answer C: The two quantities are equal.
Regrouping gives $(ab)(bc)(ca) = a^2b^2c^2 = (abc)^2$. Since $abc = -1$, its square is 1.

110. HSPT-Q12-050 - Answer D: The relationship cannot be determined from the information given.
A plane through the center gives radius 6. A plane near the surface can cut a circle of radius 2, below 3. The location is unspecified.

111. HSPT-Q12-051 - Answer A: Quantity A is greater.
Equal areas give $(1/2)b \cdot h = b \cdot h$. Canceling the shared positive height shows that the triangle's base is twice the parallelogram's.

112. HSPT-Q12-052 - Answer D: The relationship cannot be determined from the information given.
An inscribed square has area 2. An inscribed rectangle with sides 1/2 and $\sqrt{15}/2$ has diagonal 2 and area $\sqrt{15}/4 < 1$, giving the opposite comparison.

Reading Comprehension

113. HSPT-R12-001 - Answer C: buy shoes
The opening paragraph says they are supposed to buy shoes before the shops close, which explains the narrator's initial impatience.

114. HSPT-R12-002 - Answer B: Seeing a familiar person in another role can reveal abilities one has never fully noticed.
The narrator already knows Mom's occupation but newly sees her authority and competence, then recognizes the same person in a familiar gesture.

115. HSPT-R12-003 - Answer D: Her coworkers rely on her judgment and respond closely to her direction.
Coworkers wait for her words, anticipate a needed stand, and bring a card for approval, supporting an inference of professional trust.

116. HSPT-R12-004 - Answer A: brief and sharply delivered
The words are paired with exactness and contrast with her interrupted home stories, indicating a concise, sharply delivered manner.

117. HSPT-R12-005 - Answer C: The linked gestures bring her professional competence and familiar parental care into the same picture.
The same hand performs the technical and caring gestures, joining the newly observed professional role with the mother the narrator already knows.

118. HSPT-R12-006 - Answer D: To explain how a rotating wheel temporarily stores energy and smooths uneven inputs
The passage explains storage, release, mass distribution, and friction to show both what a flywheel does and what it cannot do.

119. HSPT-R12-007 - Answer A: It decreases.
The second paragraph directly states that the wheel's speed decreases as it releases energy to help the connected mechanism move.

120. HSPT-R12-008 - Answer D: More mass placed farther from the axle
The third paragraph identifies mass farther from the axle as the arrangement that increases resistance to changes in rotational speed.

121. HSPT-R12-009 - Answer C: a supply held for later use
The turning wheel draws on previously stored energy between inputs, so reserve refers to an available supply kept for later use.

122. HSPT-R12-010 - Answer A: The wheel with less friction should keep moving longer before stopping.
With the same initial store, a lower rate of energy loss should delay the point at which useful rotation ends.

123. HSPT-R12-011 - Answer C: The fishing crew returns before a storm.
The second paragraph explicitly identifies the crew's return before the storm as part of the plot that every teller must preserve.

124. HSPT-R12-012 - Answer A: Whether the differences are permitted variations under the tradition's rules
The tradition allows invented jokes, so differences alone do not establish error; the researcher must distinguish permitted invention from changes to fixed elements.

125. HSPT-R12-013 - Answer B: an added imaginative detail
The examples involve freely changing meals, descriptions, and jokes, so embellishment means an imaginative addition rather than an alteration required by the fixed plot.

126. HSPT-R12-014 - Answer B: It provides an organizing structure while leaving room for individual expression.
A spine supports movement, while a cage restricts it; the contrast presents the plot as a supporting structure rather than a complete constraint.

127. HSPT-R12-015 - Answer B: To introduce a familiar standard of sameness before showing why a different standard may fit oral performance
The opening supplies word-for-word agreement as a familiar expectation, then contrasts it with a tradition that permits controlled variation.

128. HSPT-R12-016 - Answer D: An oral tradition can preserve a shared plot while allowing variations that follow its own boundaries.
The passage combines a preserved core with permitted invention, while rejecting both strict word-for-word sameness and unlimited alteration.

129. HSPT-R12-017 - Answer B: The narrator's arm hid the moment from the coach.
The passage states that the narrator's arm hid the moment, explaining why the coach needs the narrator's report.

130. HSPT-R12-018 - Answer D: An honest account can matter more than preserving an impressive record.
The narrator gives up the clean record to report the felt touch accurately, then experiences relief without receiving any special praise.

131. HSPT-R12-019 - Answer B: He treats truthful reporting as a normal part of the drill.
Jalen responds exactly as he did before other exchanges, suggesting that honesty fits his ordinary expectations rather than requiring a ceremony.

132. HSPT-R12-020 - Answer C: It holds attention on the brief pause in which the narrator must choose an answer.
The suspended pencil makes the waiting moment visible, emphasizing the narrator's choice before the result becomes a written record.

133. HSPT-R12-021 - Answer A: The narrator feels relief from the burden of possibly concealing the truth.
The unchanged metal rules out literal weight loss; the lighter feeling expresses the relief that follows an honest answer.

134. HSPT-R12-022 - Answer C: It damages structures that kept some materials separated.
The second paragraph explains that cutting damages internal structures, allowing previously separated materials to meet while exposing the tissue to oxygen.

135. HSPT-R12-023 - Answer B: oxygen
The passage specifically identifies oxygen from the air as a participant in the reactions that lead to brown pigments.

136. HSPT-R12-024 - Answer C: To explain how several interacting conditions produce browning and how changing those conditions can affect it
The passage explains the interacting materials and conditions, then distinguishes ways those conditions can be altered and the limits of those changes.

137. HSPT-R12-025 - Answer D: something that blocks or limits passage
The coating limits oxygen reaching the surface, so barrier means something that restricts passage rather than a new reaction ingredient.

138. HSPT-R12-026 - Answer A: The pale result proves that the apple's enzymes have been destroyed.
A barrier can restrict oxygen without removing enzymes, so the absence of browning does not establish that those enzymes have been destroyed.

139. HSPT-R12-027 - Answer D: To distinguish costs of an existing sailing from costs that change when the service expands
The author separates fixed, variable, and average costs, then shows why changing from one sailing to two changes the analysis.

140. HSPT-R12-028 - Answer D: the dock fee for that sailing
The first paragraph explicitly treats the dock fee as fixed for the scheduled sailing, unlike the additional ticket-printing cost.

141. HSPT-R12-029 - Answer A: The second boat introduces additional crew and dock costs that the existing-seat decision does not add.
The added passenger uses existing capacity, but the second boat adds substantial costs; income exceeding the first decision's extra cost may not cover the second's.

142. HSPT-R12-030 - Answer C: allocated across
The paragraph discusses cost per passenger, so spread means allocated across the paying passengers rather than physically moved or removed.

143. HSPT-R12-031 - Answer A: It changes the decision from filling an existing boat to adding a sailing, showing why the cost boundary matters.
The final paragraph expands the unit of decision, revealing costs that stay unchanged within one sailing but rise when another is added.

144. HSPT-R12-032 - Answer B: Analytical, because it distinguishes costs under explicitly stated assumptions
The simplified model, careful distinctions, and repeated limits create an analytical tone rather than an emotional celebration or warning.

145. HSPT-R12-033 - Answer D: Clip it to her clothing.
Sada clips the badge to her coat before her voice carries over the stalls, establishing how the rented object operates.

146. HSPT-R12-034 - Answer B: Being heard by everyone can interfere with communicating appropriately with one person.
The badge wins public attention but prevents the privacy Orin requests; removing it lets Sada attend to a particular person.

147. HSPT-R12-035 - Answer A: Her amplified reply has drawn attention to a conversation he wanted to keep private.
Orin first requests privacy, then stops when faces appear; his gestures suggest withdrawal after Sada's voice defeats the sheltered setting.

148. HSPT-R12-036 - Answer B: moved back
The phrase a few steps shows physical movement away from Sada; receded therefore means moved back rather than approached or spoke.

149. HSPT-R12-037 - Answer D: It emphasizes the irony that fulfilling the promise has created the opposite of the connection she now needs.
The promise is literally fulfilled, yet its success obstructs a private exchange; this mismatch between expected benefit and actual consequence creates irony.

150. HSPT-R12-038 - Answer D: The loose zipper pull strikes the water flask.
The lantern reveals the metal pull swinging against the flask, and the next gust reproduces the tapping while the narrator watches.

151. HSPT-R12-039 - Answer A: Fear gives an uncertain sound a threatening meaning until observation supplies a simpler explanation.
The narrator imagines an approaching presence, then sees an ordinary cause; the unchanged landscape becomes less frightening when the sound is understood.

152. HSPT-R12-040 - Answer C: The narrator fears that showing a light could reveal a position to a supposed intruder.
The narrator imagines that brightness could announce a location, showing fear of being detected before the ordinary source of the sound is discovered.

153. HSPT-R12-041 - Answer C: put into a sheltered place
Putting the metal pull inside the boot keeps it from swinging against the flask; tucked describes placing it within that sheltered space.

154. HSPT-R12-042 - Answer B: It gives a simple possibility that the frightened narrator initially dismisses, delaying the reader's confirmation.
Aven offers the ordinary cause, but the narrator finds it inadequate; delaying visual confirmation sustains tension while exposing the gap between explanation and belief.

155. HSPT-R12-043 - Answer C: Each piece retains both kinds of pole, and the new faces attract when kept in their original orientation.
The first and third paragraphs specify two poles per piece and opposite poles at the new faces when their original orientation is preserved.

156. HSPT-R12-044 - Answer D: To explain why dividing a magnet does not separate its two kinds of pole into different pieces
The domain model explains why each remaining piece behaves as a two-pole magnet, answering the mistaken expectation raised at the beginning.

157. HSPT-R12-045 - Answer A: The facing ends would repel because they would now be like poles.
A piece's outer and cut ends have opposite poles; substituting its outer end changes unlike facing poles to like poles, which repel.

158. HSPT-R12-046 - Answer B: They raise an apparently plausible prediction, reject it, and use a model of smaller regions to explain the actual outcome.
The opening contrasts expectation with outcome; the following paragraphs supply domains as the explanatory model connecting the intact bar and its pieces.

159. HSPT-R12-047 - Answer A: Magnetic regions can remain active while their differing directions reduce the object's overall effect.
The final paragraph separates local magnetic behavior from the combined effect, which can be weakened by domains pointing in different directions.

160. HSPT-R12-048 - Answer C: The button is too heavy for the new cloth.
The remark about being too heavy occurs as the coat is remade, immediately before the thread is cut and the button is removed.

161. HSPT-R12-049 - Answer C: high in the air
The button hangs against the paper sky above the ship, so aloft describes its raised position rather than a way of fastening the coat.

162. HSPT-R12-050 - Answer A: She wants viewers to see the moon without noticing how it is suspended.
The concealed support and visible shining circle help the button appear to belong in the paper sky rather than advertise its suspension.

163. HSPT-R12-051 - Answer D: The button thinks about its usefulness and watches the ship with interest.
The button receives thoughts and feelings normally associated with a person; the other details describe ordinary actions or physical materials.

164. HSPT-R12-052 - Answer C: The narrator's role has shifted from practical fastening to imaginative participation.
Keeping closed recalls the coat's fastening; opening a story links the same object to a new imaginative role without changing its material.

165. HSPT-R12-053 - Answer B: the wire loop's rim
The third paragraph explicitly says the rim fixes the film's edge, distinguishing the loop's boundary from the bubble's enclosed-volume condition.

166. HSPT-R12-054 - Answer A: The same tendency to reduce surface area can produce different shapes under different restrictions.
The passage connects both shapes to reduced surface energy while distinguishing the fixed rim from the enclosed-volume condition.

167. HSPT-R12-055 - Answer B: The loop fixes a boundary, while the bubble comparison concerns shapes enclosing the same volume.
The proposed counterexample changes the relevant restriction; a boundary-spanning film and a volume-enclosing bubble need not share the same shape.

168. HSPT-R12-056 - Answer B: stretch across
The film forms a sheet across the loop's opening; span describes covering that bounded space rather than measuring time or breaking the rim.

169. HSPT-R12-057 - Answer D: It presents a shared physical tendency, applies it under two restrictions, and then notes complications in real examples.
The second paragraph introduces surface tension, the third distinguishes constraints, and the last acknowledges real effects while preserving the model's usefulness.

170. HSPT-R12-058 - Answer B: calling the listener at noon
The quoted promise specifies a call at noon; the other actions either belong to the listener or are not promised in the example.

171. HSPT-R12-059 - Answer D: It creates a responsibility and a basis for another person's planning, rather than merely predicting an event.
The passage distinguishes commitment from prediction, then follows the resulting responsibility and its practical consequences for the listener.

172. HSPT-R12-060 - Answer D: A commitment establishes responsibility even when outside circumstances prevent the intended result.
The passage separates undertaking responsibility from guaranteeing success; reasonable effort and explanation can remain relevant despite an obstacle.

173. HSPT-R12-061 - Answer C: dependence on a commitment
The listener plans around the speaker's commitment, so reliance means depending on it rather than controlling every circumstance or owning equipment.

174. HSPT-R12-062 - Answer A: It completes the explanation by showing the commitment's consequences from the listener's side.
The conclusion returns to the listener's planning, completing the earlier distinction between describing an event and creating a responsibility toward someone.

Mathematics

175. HSPT-M12-001 - Answer C: 7.2 × 10□□
Moving the decimal four places right gives 7.2; the factor 10□□ compensates for that change and restores the original value.

176. HSPT-M12-002 - Answer D: It is rational and lies strictly between them.
Adding rational numbers and dividing by 2 gives a rational number; averaging two different values places the result strictly between them.

177. HSPT-M12-003 - Answer C: 162
The parentheses make the first value 81; in the second, exponentiation comes before the leading minus, giving −81, so the difference is 162.

178. HSPT-M12-004 - Answer A: It neither terminates nor repeats a fixed block forever.
An irrational number has a nonterminating, nonrepeating decimal; terminating and eventually repeating decimals represent rational numbers instead.

179. HSPT-M12-005 - Answer D: $x < -5$
Dividing by −4 gives −5 and reverses the inequality direction, so the equivalent condition is $x < -5$.

180. HSPT-M12-006 - Answer B: $a = 4$; $b = 3$
Matching coefficients gives $a + b = 7$ and $2a - b = 5$; adding yields $3a = 12$, so $a = 4$ and $b = 3$.

181. HSPT-M12-007 - Answer A: $|x - 1| = 3$
Both −2 and 4 lie 3 units from 1, so $|x - 1| = 3$ has precisely those two solutions.

182. HSPT-M12-008 - Answer A: 0
Expanding gives $f(x + 1) = x^2 + x$; equality with $x^2 - x$ requires $-x = x$, so $x = 0$.

183. HSPT-M12-009 - Answer D: 33π square units
Subtract the smaller disk's area from the larger: $\pi \times 7^2 - \pi \times 4^2 = (49 - 16)\pi = 33\pi$ square units.

184. HSPT-M12-010 - Answer B: R
The four squares in the row wrap around the cube's sides; P and R face opposite directions, while T and U form the remaining opposite pair.

185. HSPT-M12-011 - Answer C: 120°
In triangle BIC, the bisected angles measure 25° and 35°; therefore angle BIC = 180° − 25° − 35° = 120°.

186. HSPT-M12-012 - Answer B: 25
Reading 70 is halfway between 20 and 120, so it corresponds to the true value halfway between 0 and 50: 25.

187. HSPT-M12-013 - Answer A: 1 millimeter
One millimeter is 0.001 meter, requiring three decimal places; the other lengths need two decimal places or none.

188. HSPT-M12-014 - Answer D: They suggest a ratio of 2 to 1, while the actual values have a ratio of 1.1 to 1.
The visible heights give 20/10 = 2, whereas the values give 110/100 = 1.1; subtracting the baseline changes the ratio.

189. HSPT-M12-015 - Answer B: 5/8
Weight the two envelope probabilities equally: (1/2)(1/2) + (1/2)(3/4) = 1/4 + 3/8 = 5/8.

190. HSPT-M12-016 - Answer C: The two recorded quantities tend to be higher together.
Association describes how the recorded quantities vary together; it does not by itself establish a causal direction or rule out other factors.

191. HSPT-M12-017 - Answer B: 3,004,000
Three million contributes 3,000,000 and four thousand contributes 4,000; adding them gives 3,004,000, with zeros preserving the unused places.

192. HSPT-M12-018 - Answer C: 0.09
Since $9^2 = 81$ and $0.09 \times 0.09 = 0.0081$, the required positive number is 0.09.

193. HSPT-M12-019 - Answer A: r/r
The quotient r/r is always 1; adding 1 or multiplying by 2 preserves irrationality, while squaring an irrational need not produce a rational number.

194. HSPT-M12-020 - Answer B: $(x - 3)(x + 3)$
Multiplying the conjugate factors gives $x^2 + 3x - 3x - 9 = x^2 - 9$, with the middle terms canceling.

195. HSPT-M12-021 - Answer C: $y = -x/2 + 3$
The required slope is −1/2; substituting (4, 1) into $y = -x/2 + b$ gives $1 = -2 + b$, so $b = 3$.

196. HSPT-M12-022 - Answer A: y = 7x
Direct proportion has the form y = kx with a constant multiplier and no added constant; y = 7x fits that form.

197. HSPT-M12-023 - Answer D: 4
For negative x the left side is 0; for nonnegative x it is 2x, so only x = 4 gives 8.

198. HSPT-M12-024 - Answer D: 56 square units
Each corner triangle has legs 4 and 7 and area 14; subtracting four such triangles from 112 leaves 56 square units.

199. HSPT-M12-025 - Answer D: 13 units
The base diagonal is $\sqrt{(3^2 + 4^2)} = 5$; the space diagonal is then $\sqrt{(5^2 + 12^2)} = 13$ units.

200. HSPT-M12-026 - Answer B: 72°
Its five equally spaced vertices divide a full turn into five matching positions, so the smallest positive rotation is 360°/5 = 72°.

201. HSPT-M12-027 - Answer C: perimeter
The strip follows the complete boundary, so its length is determined by perimeter rather than the flap's area, volume, or mass.

202. HSPT-M12-028 - Answer A: 90 seconds
The faster pointer gains 6 − 2 = 4° each second; gaining one complete turn takes 360/4 = 90 seconds.

203. HSPT-M12-029 - Answer B: 201.4 centimeters
Eight pieces require seven cuts. Their usable length is 200 centimeters; seven 2-millimeter losses add 1.4 centimeters, giving 201.4 centimeters.

204. HSPT-M12-030 - Answer A: 2 points
Using the supplied definition gives 8 × 1/4 + 0 × 3/4 = 2 points, the probability-weighted average score.

205. HSPT-M12-031 - Answer D: −1
Adding any one value leaves the median at 3; equal mean then requires a total of 15, so the added value is 15 − 16 = −1, preserving the unique mode.

206. HSPT-M12-032 - Answer C: 1/3
Only A-B-C and C-A-B put A immediately left of B, so the probability is 2 out of 6, or 1/3.

207. HSPT-M12-033 - Answer D: 255
Pairing terms or adding successively gives 255; equivalently, doubling the sum and subtracting the original leaves 256 − 1.

208. HSPT-M12-034 - Answer A: −50
Each consecutive pair, such as 1 − 2 or 99 − 100, equals −1; fifty pairs therefore total −50.

209. HSPT-M12-035 - Answer C: It is a perfect square.
Factors normally pair as d and n/d; an odd count requires one unpaired factor equal to its partner, so n is a square.

210. HSPT-M12-036 - Answer A: ac/b
Dividing by b/c means multiplying by c/b, so a ÷ (b ÷ c) = a × c/b = ac/b.

211. HSPT-M12-037 - Answer C: {(1, 4), (2, 4), (3, 4)}
The correct set assigns one output to each distinct input; repeating an output is allowed, but assigning two outputs to one input is not.

212. HSPT-M12-038 - Answer A: x > 5 or x < −3
The first inequality gives x > 5, and the second gives x < −3; OR includes either separate solution interval.

213. HSPT-M12-039 - Answer C: Only x = 4
For x = −1, the original equation reads 2 = −2, which is false; for x = 4, it reads 3 = 3.

214. HSPT-M12-040 - Answer B: 6
With x = 6, the rule gives f(f(6)) = 6; substituting the known value f(6) = 11 gives f(11) = 6.

215. HSPT-M12-041 - Answer C: 6π + 12 units
The curved edge is half a circumference, 6π; the straight edge is the diameter, 12, giving a total of 6π + 12 units.

216. HSPT-M12-042 - Answer D: trapezoid
A trapezoid can have just one parallel pair; every square, rectangle, and parallelogram has two pairs of parallel sides.

217. HSPT-M12-043 - Answer B: (7, −7)
The displacement is (3 − 1, −1 − 2) = (2, −3). Tripling gives (6, −9); adding (1, 2) gives (7, −7).

218. HSPT-M12-044 - Answer D: 50 milliliters
If x milliliters are added, pigment balance gives 20 + 0.50x = 0.30(100 + x), so 0.20x = 10 and x = 50.

219. HSPT-M12-045 - Answer A: 2000
Dividing 2000 by 400 gives the integer 5; the other listed century years are not divisible by 400 and do not qualify.

220. HSPT-M12-046 - Answer D: A has span 4, and B has span 9.
Subtract the quartile endpoints: A has 14 − 10 = 4, while B has 17 − 8 = 9, a wider middle-half span.

221. HSPT-M12-047 - Answer B: 7/9
There are 27 equally likely sequences. Only 3 × 2 × 1 = 6 have all symbols different, leaving 21/27 = 7/9 with a match.

222. HSPT-M12-048 - Answer B: 8
A unique mode must occur more often than either other category; medium therefore needs more than 7 jackets, making 8 the minimum.

223. HSPT-M12-049 - Answer B: Neither prime nor composite, because it has exactly one positive factor
The only positive factor of 1 is 1, so its factor count meets neither the prime definition nor the composite definition.

224. HSPT-M12-050 - Answer D: 4/9
The positive square is (3/2) × (3/2) = 9/4; its reciprocal is 4/9. A negative exponent does not make the value negative.

225. HSPT-M12-051 - Answer D: 5/12
An intervening p/q requires 7p/3 < q < 5p/2. Numerators 1–4 allow no integer q; p = 5 allows q = 12. Larger numerators require denominators above 12.

226. HSPT-M12-052 - Answer B: the postage returned
The postage is the output determined by the input mass, making postage the dependent variable in the stated relationship.

227. HSPT-M12-053 - Answer B: x = −2
Every point on a vertical line has the same x-coordinate. Since the given point has x = −2, the line is x = −2.

228. HSPT-M12-054 - Answer A: 3/2
Tripling x multiplies x^2 by 9, while multiplying z by 6 divides the result by 6; the net factor is 9/6 = 3/2.

229. HSPT-M12-055 - Answer D: $y = (x - 3)^2 - 2$
The original vertex (0, 0) moves to (3, −2). Using h = 3 and k = −2 gives $y = (x - 3)^2 - 2$.

230. HSPT-M12-056 - Answer A: a circle centered at O
A circle consists of points at one fixed distance from its center; a filled disk also includes closer points and therefore does not qualify.

231. HSPT-M12-057 - Answer C: 5 units
The semicircle's curved edge is 10π. Setting the base circumference 2πr equal to 10π gives r = 5 units.

232. HSPT-M12-058 - Answer A: $2\sqrt{5}$ units
The unfolded rectangle measures 4 by 2 units. Its diagonal is $\sqrt{(4^2 + 2^2)} = \sqrt{20} = 2\sqrt{5}$, unlike the interior diagonal or an edge-only route.

233. HSPT-M12-059 - Answer C: 127.5 seconds
The display means 2 minutes plus 7.50 seconds; converting the minutes gives 120 + 7.50 = 127.5 seconds.

234. HSPT-M12-060 - Answer D: Cart B uses 4 liters less.
Cart A uses 3 × 8 = 24 liters. Cart B uses 300 ÷ 15 = 20 liters, so B uses 4 liters less.

235. HSPT-M12-061 - Answer A: 112 through 120 grams
The minimum is (164 − 2) − (48 + 2) = 112; the maximum is (164 + 2) − (48 − 2) = 120 grams.

236. HSPT-M12-062 - Answer C: 10/3
The mean is 12. Absolute deviations are 4, 1, and 5; their average is (4 + 1 + 5)/3 = 10/3.

237. HSPT-M12-063 - Answer B: 2–5 minutes; 3 more
Multiply width by height: the first frequency is 2 × 6 = 12, and the second is 3 × 5 = 15, exceeding it by 3.

238. HSPT-M12-064 - Answer C: 0
If three cards occupy their matching places, the sole remaining card must match the sole remaining place. Exactly three matches are impossible, so the probability is 0.

Language

239. HSPT-L12-001 - Answer C: we'll
The apostrophe in we'll marks the omitted letters from will, leaving we plus ll as the contracted form.

240. HSPT-L12-002 - Answer D: Dear Director Patel:
A formal business-letter salutation conventionally ends with a colon, distinguishing it from ordinary sentences and questions.

241. HSPT-L12-003 - Answer A: the brothers-in-law's workbenches
The compound noun becomes plural at brothers, but its possessive ending attaches to the end of the entire compound: brothers-in-law's.

242. HSPT-L12-004 - Answer B: We used a Nikon camera to photograph the dunes.
Nikon is a brand name and takes a capital letter; camera and dunes are common nouns in this sentence.

243. HSPT-L12-005 - Answer D: She enrolled in Oceanography 205 after taking introductory oceanography.
The numbered course title Oceanography 205 is capitalized, while the general subject oceanography remains lowercase.

244. HSPT-L12-006 - Answer B: dilemma
Dilemma contains one l followed later by a double m; none of the other spellings preserves that standard arrangement.

245. HSPT-L12-007 - Answer D: hierarchy
Hierarchy begins with hier and includes the sequence archy; the alternatives reverse the opening vowels or omit a required r.

246. HSPT-L12-008 - Answer C: whose
Whose marks the relationship between the telescope and its lens; this possessive relative pronoun can refer to an inanimate object.

247. HSPT-L12-009 - Answer C: repaired
In the construction had an object acted on, the past participle follows the object: had the axle repaired.

248. HSPT-L12-010 - Answer A: had the door closed
Beginning with no sooner requires auxiliary-subject inversion, so had precedes the subject the door in the completed sentence.

249. HSPT-L12-011 - Answer B: assure; ensure
Assure addresses a person's concern, while ensure means make certain that a condition is satisfied, here the gate's being closed.

250. HSPT-L12-012 - Answer C: ingenious; ingenuous
Ingenious describes clever invention, while ingenuous describes innocent, straightforward openness, matching the two contexts respectively.

251. HSPT-L12-013 - Answer D: (2, 5), then (3, 7), then (4, 6)
Sentences 2 and 5 explain joining pages, 3 and 7 address opening, and 4 and 6 discuss production.

252. HSPT-L12-014 - Answer A: Thread binding stitches folded page groups together, whereas adhesive binding glues prepared page edges into the cover.
Whereas presents the contrast while retaining which method uses stitches and which uses glue, without reversing or merging the processes.

253. HSPT-L12-015 - Answer A: Both methods attach pages to a cover, but they differ in construction, opening behavior, and production.
The addition identifies the common purpose and previews the three comparison features that organize the following descriptions.

254. HSPT-L12-016 - Answer D: "We waited ... before the gate opened."
An ellipsis marks words omitted from a quotation; the three dots replace through the cold afternoon without changing the remaining word order.

255. HSPT-L12-017 - Answer A: The pilot Lena Cho will speak first.
Because the name identifies which pilot among several is meant, it is essential information and should not be set off by commas.

256. HSPT-L12-018 - Answer C: Artists of the Renaissance studied proportion.
Renaissance names a historical era and is capitalized; studied and proportion remain ordinary lowercase words in this sentence.

257. HSPT-L12-019 - Answer D: The harbor tug uses a diesel engine.
The common noun diesel is lowercase when naming an engine type, even though it originated from a person's surname.

258. HSPT-L12-020 - Answer C: vacuum
Vacuum contains one c and two consecutive u's; its spelling differs from forms that double c or omit a u.

259. HSPT-L12-021 - Answer D: rendezvous
Rendezvous includes the z before vous and ends in s; the alternatives omit that z or change the ending.

260. HSPT-L12-022 - Answer B: liaison
Liaison has the letter sequence l-i-a-i-s-o-n, retaining both i's in the correct positions rather than omitting or moving one.

261. HSPT-L12-023 - Answer D: The potter herself signed the base.
In the correct sentence, herself emphasizes the potter and can be removed; the base, rather than herself, receives the action.

262. HSPT-L12-024 - Answer B: An; a
Honest begins with a vowel sound because h is silent; useful begins with a consonant y sound, requiring an and a respectively.

263. HSPT-L12-025 - Answer A: swim; swimming
Used to takes a base verb for a former habit, while is used to takes a gerund to describe being accustomed to something.

264. HSPT-L12-026 - Answer A: adopt; adapt
Adopt means accept formally or take as one's own, while adapt means modify something to suit a particular use or condition.

265. HSPT-L12-027 - Answer A: precede; proceed
Precede means come before, fitting the introduction; proceed means move on or continue, fitting the transition to questions.

266. HSPT-L12-028 - Answer C: Demonstration Moved to October 21 in the North Room
The date and room change, while the time and confirmed reservations remain unchanged; the heading accurately highlights the two updates.

267. HSPT-L12-029 - Answer B: Please cancel your old reservations and book again before October 18.
The notice preserves confirmed places without further action, so instructing those guests to cancel and rebook directly contradicts its purpose.

268. HSPT-L12-030 - Answer B: You may request a place before October 18 for the October 21 demonstration in the North Room at 4 p.m.
The reply preserves the request deadline, revised date and room, unchanged time, and the distinction between requesting a place and already having one.

269. HSPT-L12-031 - Answer D: 1,234,567
American notation groups whole-number digits in threes from the right; commas therefore separate 1 from 234 and 567.

270. HSPT-L12-032 - Answer C: "She [Dr. Miro] returned before sunset."
Square brackets identify wording supplied by an editor inside a quotation, distinguishing the clarification from the original speaker's words.

271. HSPT-L12-033 - Answer A: The casting had one serious flaw: a crack ran through its base.
The complete first clause identifies a flaw; the colon introduces the second clause explaining it, without splitting a verb from its object.

272. HSPT-L12-034 - Answer A: Canis lupus
Canis begins the scientific name and is capitalized; lupus is the second word and remains lowercase under the stated rule.

273. HSPT-L12-035 - Answer D: She holds a Ph.D. in geology.
The conventional abbreviation Ph.D. capitalizes P and D while leaving h lowercase; the ordinary field name geology stays lowercase.

274. HSPT-L12-036 - Answer C: cemetery
Cemetery uses e in each of its first three syllables and one r; the alternatives replace, omit, or double a letter.

275. HSPT-L12-037 - Answer B: broccoli
Broccoli contains two c letters and one l; the alternatives omit a c or incorrectly double the l.

276. HSPT-L12-038 - Answer B: haven't they
An affirmative clause with have takes the negative tag haven't, and the plural subject gates is represented by they.

277. HSPT-L12-039 - Answer A: It
It serves as the introductory subject for the following that clause; the sentence does not identify a plural group or introduce a noun after there.

278. HSPT-L12-040 - Answer A: The rope that we bought is strong.
In the correct sentence, we supplies the relative clause's subject, so object that can disappear; the other clauses need that as their subject.

279. HSPT-L12-041 - Answer C: adverse; averse
Adverse means unfavorable, describing the weather; averse means opposed or unwilling, describing the crew's attitude toward postponement.

280. HSPT-L12-042 - Answer B: discreet; discrete
Discreet describes tactful or unobtrusive behavior, while discrete means distinct or separate; the inquiry and the sections require those different meanings.

281. HSPT-L12-043 - Answer D: the tunnel; the viaduct
Former points to the first of two previously mentioned things, while latter points to the second, preserving the report's stated order.

282. HSPT-L12-044 - Answer C: A cookbook might describe how dough is folded, while a chef's profile might identify the person in the same photograph.
The new example applies the same-image, different-purpose principle to another subject; the other choices ignore purpose or prescribe irrelevant priorities.

283. HSPT-L12-045 - Answer B: The chisel rests on the stone's upper edge, with the mason holding the hammer above it.
The correct description adds visible tool placement relevant to the article; the alternatives repeat the caption, emphasize background, or make an unsupported judgment.

284. HSPT-L12-046 - Answer D: .
The sentence reports a question rather than directly asking it; its main clause is declarative and therefore ends with a period.

285. HSPT-L12-047 - Answer D: Rina said, "The sign reads 'Keep clear.'"
The introductory comma precedes the outer quotation, the inner quotation closes first, and the final period goes inside both closing quotation marks.

286. HSPT-L12-048 - Answer C: The diagram is ready, Your Honor.
Your Honor is a formal title used in direct address, so both words are capitalized when the witness addresses the judge.

287. HSPT-L12-049 - Answer D: The reviewer called the final movement "a quiet farewell."
An integrated quoted phrase keeps its ordinary lowercase beginning; quotation marks alone do not turn it into a new sentence or title.

288. HSPT-L12-050 - Answer C: porcelain
Porcelain ends with the letter sequence l-a-i-n; the alternatives omit the a, add an n, or replace the final letters.

289. HSPT-L12-051 - Answer D: turbulence
Turbulence has u after turb and the ending lence; the alternatives replace a vowel or add an extra final e.

290. HSPT-L12-052 - Answer B: supersede
The replacement verb uses the ending sede; substituting cede or ceed follows a tempting but incorrect spelling pattern.

291. HSPT-L12-053 - Answer C: confusing; confused
Confusing describes what causes confusion; confused describes the people experiencing it, so the participial adjectives require different endings.

292. HSPT-L12-054 - Answer B: The divers checked each other's clips.
Each other expresses the reciprocal action, and its possessive form is each other's; their own would change the action to checking personal equipment.

293. HSPT-L12-055 - Answer C: arrives
A future-time clause introduced by when normally uses the present tense; the main clause's will begin already locates the event in the future.

294. HSPT-L12-056 - Answer A: The visitor was warned not to flout the rules.
Flout means openly disregard a rule, while flaunt means display something conspicuously; the other sentences exchange those meanings.

295. HSPT-L12-057 - Answer A: elicit
Elicit means draw out a response; illicit means unlawful, emit means send out, and omit means leave out, none matching the counselor's purpose.

296. HSPT-L12-058 - Answer A: beside
Beside means next to, describing the table's position; besides means in addition to, while between and through do not fit the stated arrangement.

297. HSPT-L12-059 - Answer B: A familiar name can acquire a new use through one important resemblance.
The proposed topic introduces the transfer of a familiar name through a shared feature, preparing both the digital example and the comparison's limits.

298. HSPT-L12-060 - Answer A: Omit it because it turns a limited resemblance into an unsupported claim about all features. The passage bases the borrowed name on a framed view while preserving other differences; the proposed sentence wrongly extends the resemblance to every feature.